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Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Acronyms

DDS	Data Distribution Service
EMSA	European Maritime Safety Agency
LTI	linear time-invariant
MPC	Model Predictive Control
NMPC	Non-linear Model Predictive Control
PID	proportional–integral–derivative
ROS	Robot Operating System
SITL	Software-in-the-Loop
UAV	Unmanned Aerial Vehicle
USV	Unmanned Surface Vessel
WAM-V	Wave Adaptive Modular Vessel

Chapter 1

Introduction

This chapter introduces the thesis, presenting the context, motivation, and objectives of the research, followed by an overview of the system architecture, the expected scientific and engineering contributions, and the overall structure of the dissertation.

1.1 Motivation

Saving lives at sea, protecting marine ecosystems, and securing maritime infrastructure all rely on one essential capability: understanding what is happening, where, and when. In vast and dynamic maritime environments, this requires maintaining reliable, high-resolution situational awareness over long periods. The European Maritime Safety Agency (EMSA) recognises this challenge and highlights the need for continuous monitoring and locally persistent operations to enable effective maritime surveillance and response [1].

Different maritime assets meet this challenge in complementary ways. Surface vessels give a physical presence at sea, with long endurance and large payload capacity [2, 3]. However, their range is limited by line of sight, and operations are expensive [4]. Aerial platforms, such as manned aircraft and Unmanned Aerial Vehicles (UAVs), allow fast deployment and elevated sensing [5]. Yet, aircraft are costly, and UAVs suffer from low endurance and unreliable communication [6].

Together, these assets form a connected system rather than separate solutions. Each one compensates for what the other lacks. Unmanned Surface Vessels (USVs) provide long-term presence at sea at much lower cost than crewed vessels, allowing wide and affordable patrols. UAVs extend this reach by sensing beyond the horizon from the air. Working as a team, they maintain continuous and local maritime awareness. This motivates the study of coordinated USV–UAV systems, where cooperation creates mission capabilities that neither platform can achieve alone in complex and changing environments.

A natural way to strengthen this cooperation is to add a physical tether between both platforms. The tether provides continuous power and reliable communication between the USV and UAV [7]. By placing the energy supply and communication systems on the USV, it overcomes two main limits of maritime

UAVs: short endurance and weak long-range communication. This approach has recently transitioned from academic concepts to real-world industrial systems; for instance, the marine robotics company Exail partnered with Elistair to integrate a tethered quadrotor directly onto the DriX USV catamaran for persistent maritime surveillance [8]. Similarly, ACUA Ocean and Teledyne FLIR Defense demonstrated an integrated tethered aerial-surface system for the UK Ministry of Defence to support long-endurance maritime security and target detection [9].

Figure 1.1 illustrates the configuration of the tethered USV–UAV system considered in this work.

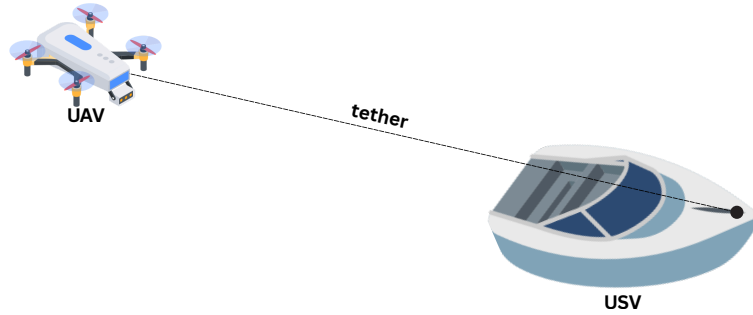


Figure 1.1: Schematic of a tethered USV–UAV system.

The main challenge of this work is to design a control framework that enables two autonomous vehicles, connected by a tether, to operate as a single coordinated system. The controller must manage not only the motion of both vehicles but also account for the tether constraints and dynamics, ensuring safe operation and robustness to wind, waves, and obstacles.

The USV and UAV operate under very different dynamics. The USV moves slowly, dominated by hydrodynamic drag and inertia, while the UAV reacts quickly, with motion evolving on much shorter time scales. Coordinated control is challenging because the controller must handle these different response rates and actuation dynamics consistently.

The maritime environment adds further complexity. Wind, waves, and water motion constantly disturb both platforms, making robustness to disturbances essential. Nearby structures, vessels, or obstacles can restrict their movement and risk tether entanglement. Effective cooperation relies on continuous information exchange and synchronised decision-making between both agents.

Most existing tethered UAV systems focus on power delivery and communication, with operation often remaining manual or semi-autonomous. In research prototypes, the aerial and surface vehicles are usually decoupled, the tether is simplified or linearised, or the surface platform is treated as fixed. These assumptions make control design easier but reduce realism. Few cooperative control approaches explicitly model the tether while coordinating both agents under realistic maritime conditions.

1.2 Problem Statement

This work addresses the problem of controlling a cooperative system composed of an autonomous surface vessel (the WAM-V USV) and a physically connected quadrotor (the x500 UAV) operating in a

coordinated manner. The objective of such a system is to autonomously execute maritime missions by following predefined waypoints or continuous reference trajectories, while respecting the physical and operational constraints of both platforms and of the tether that connects them.

To achieve this objective, the control framework must be capable of handling different phases of operation and mission requirements. This includes the definition of multiple control modes that allow the system to perform manoeuvres such as takeoff and landing, as well as different forms of coordinated motion during navigation. The UAV NMPC must actively prevent the tether from exceeding its maximum physical length, utilizing soft constraints and slacks, while the USV NMPC coordinates the planar movement of the dynamic anchor deck.

1.3 System Overview

Interaction with the system is defined at the mission level. The operator selects a mission, such as patrolling a given area, following a moving target, or visiting a set of waypoints that represent regions of interest. This mission information is received by a behaviour management layer, which selects the active control mode and handles mission logic, including switching between different relative configurations of the UAV with respect to the USV, as well as executing specific manoeuvres such as takeoff and landing.

Based on the selected mission and active control mode, the system generates coordinated motion for both the USV and the UAV. This process combines planning and control to compute trajectories and actuation commands that satisfy vehicle and tether constraints while pursuing the mission objectives. The WAM-V USV acts as the leader planar platform carrying an active winch system on its deck. The winch dynamically spools the tether in or out to manage slack and cable tension. The x500 UAV follows a relative orbit or observation trajectory above the USV, using its elevated perspective to record visual data.

Figure 1.2 presents a high-level block diagram of the proposed system, showing the interaction between the coordinator, the decentralized NMPC solvers, the vehicles, and the co-simulation interfaces.

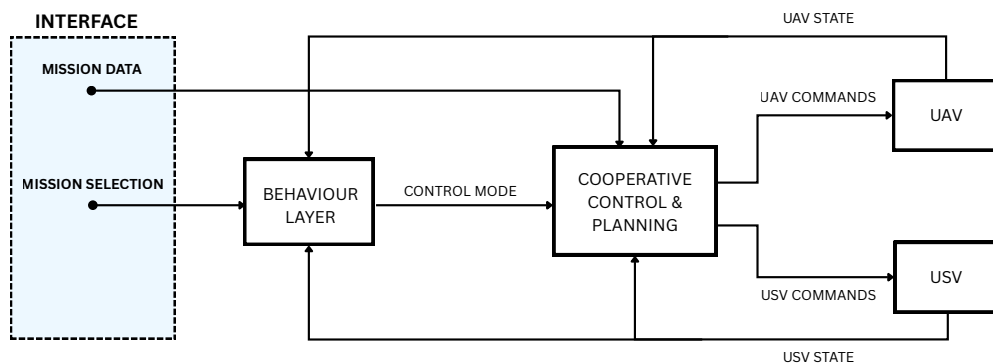


Figure 1.2: High-level block diagram of the proposed cooperative control system.

1.4 Expected Contributions

This thesis contributes to the design, implementation, and evaluation of cooperative control solutions for tethered UAV–USV systems, to enable autonomous and persistent maritime missions. The main contributions of this work are summarised as follows:

- A unified, C++ ROS 2 control pipeline utilizing Acados and HPIPM to generate and execute real-time Nonlinear Model Predictive Control solvers for both surface tracking and aerial guidance.
- The modeling and experimental identification of the UAV inner-loop attitude response lag ($\tau = 0.15$ s) using PX4 doublets flight data.
- The design and implementation of a virtual winch controller operating in both geometric and tension-feedback PD modes to prevent cable slackness and tension spikes.
- A realistic co-simulation setup in Gazebo Harmonic, integrated with the PX4 Software-in-the-Loop (SITL) firmware and the MoorDyn physics engine for multi-body lumped-mass line representation.
- The verification of NMPC constraints enforcement (states, inputs, and tether length limits) and evaluation of solver latency.
- The validation of the entire cooperative control stack under an integrated search and pursuit mission targeting a vessel teleoperated in real-time.

1.5 Thesis Structure

The remainder of this dissertation is organized as follows:

- **Chapter 2 (Literature Review and State of the Art)** reviews existing research in cooperative marine-aerial robotics, tethered drones, and predictive control algorithms.
- **Chapter 3 (System Modeling)** derives the kinematic and dynamic models of both vehicles, as well as the coupling tether and winch formulations.
- **Chapter 4 (Cooperative Control Design)** details the decentralized NMPC mathematical formulation, cost functions, physical constraints, and winch spooling control laws.
- **Chapter 5 (Simulation Environment and Software Implementation)** introduces the physical characteristics of the virtual WAM-V and x500 platforms, the simulation setup, and the software modularity.
- **Chapter 6 (Results and Discussion)** evaluates the model validation, parameter identification, tracking precision, constraint enforcement, solver latency, and the target pursuit mission.
- **Chapter 7 (Conclusions and Future Work)** summarizes the findings of the thesis and outlines directions for future research.

Chapter 2

State of the Art and Related Work

This chapter is structured into four main sections, each representing a key element of cooperative system control. Each section presents the main techniques commonly used in control theory, together with subsections reviewing related work on topics closely related to this research.

The first section addresses *modelling and identification*, which provides the mathematical and experimental tools required to describe system dynamics and estimate relevant parameters, including tether and cable physics. The second focuses on *control strategies*, ensuring stability, accurate tracking, and robustness to disturbances at the individual agent level, including predictive control and active winch architectures. The third section concerns *cooperative coordination*, which enables communication and shared decision-making so that multiple agents can operate as a unified system. Finally, *path and trajectory planning* deals with the generation of feasible and safe motion plans that connect mission objectives with low-level control actions.

2.1 Modeling and System Identification

Understanding how a system evolves is a prerequisite for any control design. Modelling provides this understanding by offering a mathematical description of the system dynamics, capturing the dominant behaviours while deliberately neglecting secondary effects that are not essential for analysis or control purposes [10]. The level of detail of such a model is inherently linked to the available physical insight, the amount and quality of experimental data, and the specific objectives for which the model is intended.

2.1.1 White-Box, Grey-Box, and Black-Box Modeling

Depending on the available physical knowledge and data, modeling methodologies are classified into three main groups:

- **White-Box Modelling:** Based on first principles and physical laws that govern system behaviour. By expressing the dynamics through analytical formulations, often derived from Newtonian mechanics, hydrodynamics, or aerodynamics, this approach provides clear insight into stability, con-

trollability, and energy transfer. To preserve analytical tractability, simplifying assumptions such as linearisation or the neglect of higher-order effects are commonly introduced.

- **Grey-Box Modelling:** Lies between purely analytical and purely data-driven approaches. The structure of the model is defined using physical insight, while parameters that are difficult to determine analytically, such as hydrodynamic or aerodynamic coefficients, are estimated from experimental data. This combination preserves a clear physical interpretation while improving accuracy through calibration.
- **Black-Box Modelling:** Relying entirely on measured input–output data to describe system behaviour. Rather than using explicit physical laws, this approach identifies mathematical relationships that best fit the observed data. Typical examples include linear parametric models, state-space representations obtained from data, and learning-based methods.

2.1.2 Tether Dynamics Modeling

Modeling flexible tethers is critical since they introduce highly nonlinear forces and kinematic constraints on the vehicles:

- **Quasistatic Catenary Models:** Describe the steady-state profile of a heavy cable suspended between two points. These models use catenary equations to resolve cable sag and tension analytically, assuming the line is in static equilibrium at each step. While computationally efficient, they neglect dynamic effects such as transverse wave propagation and drag forces, making them unsuitable for rapid aerial maneuvers.
- **Lumped-Mass Models:** Discretize the tether into a finite chain of nodes, or "lumped masses", connected by spring-damper segments. Each segment models axial elasticity, damping, and hydrodynamic drag, while the nodes resolve gravitational and buoyancy forces. This formulation, implemented in solvers such as MoorDyn, offers a high-fidelity representation of transient tether whip, slackness, and hydrodynamic drag under dynamic boundary conditions.
- **Finite-Segment and Multi-Body Models:** Formulate the line as a chain of rigid cylinders connected by spherical joints. This approach computes bending and torsional moments accurately, but requires solving high-index differential-algebraic equations, which increases the computational burden.

2.1.3 Optimization-Based Estimation

System identification offers an alternative path by constructing models directly from measured input–output data when physical knowledge is incomplete. Optimisation-based estimation focuses on identifying model parameters offline by minimising the mismatch between measured and predicted system outputs. This is typically achieved through cost functions based on least-squares formulations, including weighted least-squares and prediction-error methods (PEM) [11]. For nonlinear or constrained problems,

advanced optimisation schemes such as trust-region methods and constrained least-squares are often employed, allowing grey-box models to be calibrated while enforcing physical bounds and regularisation.

2.1.4 Related Work on System Identification

In systems connected by a tether, the model of the overall system must account at least for the tether geometry to constrain the relative motion of the agents and avoid physically infeasible configurations that could lead to tether failure.

In [12], a buoy–UAV system is presented in which the tether is modelled explicitly as a geometric and dynamic constraint between the buoy and the aerial vehicle. The tether tension is computed from the coupled dynamics and incorporated as an external force acting along the tether direction in the UAV model. This formulation relies on simplifying assumptions, namely that the tether is inextensible, massless, free of hydrodynamic effects, and remains taut during coupled operation.

An alternative modelling perspective is proposed in [13], where tether-induced forces are not included explicitly in the system dynamics. Instead, their effect is lumped into an external disturbance acting on the UAV. This disturbance is then handled at the control level through active disturbance rejection, avoiding the need for explicit tether modelling while still compensating for its influence in real time.

In [14], a cooperative UAV–USV system is modelled for the purpose of manipulating a floating object. Both vehicles are connected to the object through individual tethers, and the full system is described using simplified white-box models. The formulation assumes inextensible and massless tethers that remain permanently taut.

In [15], a tether is not included, and both the UAV and the USV are described using simplified white-box models. The USV is represented by a planar marine vehicle model commonly used for path-following applications, while the UAV is modelled independently using a reduced-order dynamic model.

2.2 Control Strategies for Autonomous Vehicles

Control theory provides the mathematical framework for designing systems that regulate their behaviour through feedback, ensuring stability, accuracy, and robustness in the presence of uncertainty and disturbances.

2.2.1 Classical and Modern Control

Classical control is based on the representation of systems through transfer functions in the Laplace or frequency domain, under the assumption of linear time-invariant (linear time-invariant (LTI)) dynamics. Within this framework, feedback laws are typically implemented through proportional–integral–derivative (proportional–integral–derivative (PID)) controllers and frequency-domain compensators designed to shape the closed-loop response.

Modern control theory generalises these principles to systems that are multivariable, time-varying, or subject to strong coupling between states and inputs. It is formulated in the *state-space* domain, where

the internal dynamics of a system are described by sets of first-order differential equations. This representation enables systematic analysis of controllability, observability, and stability. Within this framework, optimal control theory defines feedback laws that minimise a performance index, such as the linear quadratic regulator (LQR) or robust control methods such as H_∞ [16, 17].

2.2.2 Predictive and Optimization-Based Control

Predictive and optimisation-based control methods determine control actions by solving an optimisation problem that predicts the system behaviour over a finite horizon. At each sampling instant, the controller minimises a cost function subject to system dynamics and operational constraints, typically penalising deviations from reference trajectories and excessive control effort. Only the first control action is applied, and the process repeats at the next time step, forming the receding-horizon principle that defines Model Predictive Control (MPC) [18].

Extensions to nonlinear systems led to the development of Non-linear Model Predictive Control (NMPC), which replaces linear models with nonlinear formulations and uses numerical optimisation techniques to compute control inputs in real time [19, 20]. To enable real-time embedded execution on high-rate systems (e.g. UAV flight control), modern applications rely on symbolic framework code generators (such as CasADi) that export native C solvers. These solvers utilize highly efficient quadratic programming kernels such as HPIPM and execution frameworks like the Sequential Quadratic Programming Real-Time Iteration (SQP-RTI) scheme, which performs a single optimization iteration per time step to minimize solver latency.

2.2.3 Active Winch and Tension Control

In tethered UAV operations, winch design is a core element:

- **Passive Tension Tethers:** Rely on spring-loaded spools or constant-torque slip rings. While simple and requiring no electrical control, they cannot adapt to fast vessel maneuvers, leading to large tension spikes or excessive slack.
- **Active Spooling Winches:** Utilize motorized actuators regulated by microcontrollers. The winch dynamically adjusts the tether length based on geometric tracking (distance between platforms) or force-feedback algorithms.
- **Tension-Feedback and Active Heave Compensation (AHC):** Read force telemetry from a load cell. A closed-loop proportional-derivative (PD) controller regulates the winch spooling speed to maintain a target tension. In maritime environments, this functions as AHC, isolating the flying quadrotor from wave-induced vessel heave.

2.2.4 Related Work on Tethered Vehicle Control

In tethered UAV systems, the tether introduces additional forces and coupling effects that influence control design. In [12], this coupling is addressed through a nonlinear backstepping controller derived

directly from the white-box dynamics of the UAV–buoy system. The work in [13] adopts an alternative approach based on active disturbance rejection, where the tether influence is not explicitly modelled but treated as an external disturbance estimated in real time by a reduced-order observer.

In [14], cooperative manipulation with two tethered agents is controlled using a MPC formulation based on simplified white-box models of the UAV, the USV, and the floating object. A similar predictive strategy is presented in [21], where both the UAV and the USV are coordinated using NMPC, relying on known dynamic models.

In [15], the USV employs a line-of-sight guidance law combined with PID control for heading and speed regulation. The UAV motion is controlled using an NMPC formulation, which generates reference accelerations to track the desired relative trajectory. No joint or centralised control law is employed, and each vehicle is controlled using its own dedicated controller.

2.3 Cooperative Coordination and Systems

Cooperative coordination refers to the control and organisation of multiple agents working towards achieving goals through communication and joint decision-making.

2.3.1 Centralized vs. Distributed Coordination

Depending on how information is exchanged and the control authority is structured, cooperative architectures are broadly categorised as centralized or distributed:

- **Centralized Coordination:** A single decision node aggregates global information and computes control commands for all agents. This architecture facilitates global optimisation and ensures consistency of mission-level objectives, since the planner has access to full system states and constraints. However, centralized control scales poorly with team size and is vulnerable to communication delays or single-point failures.
- **Distributed Coordination:** Eliminates the need for a central planner by allowing each agent to make decisions based on locally available information and limited communication. Agents exchange state or intent data to achieve group-level objectives. Distributed coordination provides scalability and robustness to communication losses and agent failures, though it faces challenges related to synchronization and network stability.

2.3.2 UAV-USV Cooperative Systems

Cooperative deployments of UAVs and USVs combine the speed and altitude of aerial sensing with the weight-bearing capability and range of marine hulls:

- **Autonomous Landing on Moving Surfaces:** Focuses on landing a quadrotor on a moving deck (helideck) in waves. This requires precise relative state estimation (visual markers, RTK-GPS) and

predictive control to synchronize the UAV's landing velocity with the sea-state pitch/roll profile of the USV deck.

- **Mobile Recharging Stations:** The USV acts as a mother ship. Drones fly reconnaissance sweeps, return to land on the USV deck to swap or recharge batteries, and take off again, enabling long-term persistence at sea.
- **Cooperative Target Tracking:** Cooperative path-following algorithms steer the USV planar trajectory to intercept a target, while the UAV orbits above to provide continuous visual tracking and image feedback.

2.3.3 Related Work on Maritime Cooperative Robotics

In [21], cooperation is based on a tracking strategy in which the UAV is controlled to follow and land on a moving USV. Only the UAV is actively controlled, while the USV moves independently and provides its state as a reference. The interaction is therefore unidirectional, with the UAV adapting its motion to the USV.

In contrast, [14] proposes a centralised model predictive control approach in which the UAV and the USV are controlled jointly. A single optimisation problem computes the control inputs for both vehicles, explicitly accounting for the coupling introduced by the tether. This approach focuses on cooperative object manipulation rather than on mission-level coordination between the vehicles.

In [15], coordination is achieved through a leader–follower structure. The USV acts as the leader and follows a predefined path, while the UAV adjusts its motion to maintain a desired relative configuration with respect to the USV. Coordination is therefore based on reference sharing and relative motion constraints, without joint optimisation or centralised decision-making.

2.4 Path and Trajectory Planning

Path and trajectory planning address the problem of generating feasible, collision-free, and dynamically consistent motions for autonomous systems operating in structured or unstructured environments, translating abstract goals into trackable trajectories.

2.4.1 Offline and Online Planning

Methods are broadly divided into offline and online approaches, depending on when the plan is computed and how environmental information is handled:

- **Offline Planning:** Performed before execution, assuming full or near-complete knowledge of the operating environment. Classical approaches include graph-search methods (Dijkstra's, A*), sampling-based methods (RRT, PRM), and trajectory optimization. While generating globally optimal paths, they are computationally demanding and cannot adapt to dynamic changes.

- **Online Planning:** Addresses the generation and adaptation of trajectories in real time under dynamic or uncertain conditions. Reactive techniques such as potential field methods and dynamic window approaches provide fast obstacle avoidance but can suffer from local minima. Optimization-based MPC frameworks integrate vehicle dynamics into the planning loop, generating dynamically consistent collision-free paths online.

2.4.2 Related Work on Trajectory Planning

A centralised predictive planning approach is presented in [14], where a single MPC formulation jointly optimises the motion of both the UAV and the USV. In this case, planning is achieved online through joint trajectory optimisation that explicitly accounts for the coupling introduced by tethers.

Planning is addressed from a different perspective in [15], where cooperative behaviour is achieved through coordinated path following. A predefined path is assigned to the USV, while the UAV generates its motion online as part of the control process to maintain coordination. In this framework, planning is limited to the online generation of the UAV motion, and no cooperative path planning between the two vehicles is performed.

Overall, existing works typically adopt one of two approaches to planning in UAV–USV systems, either a predefined path is assigned to one vehicle while the other generates its motion online to follow or support it, or both vehicles generate their motion online as part of a predictive control framework.

Chapter 3

System modeling

Real-time control needs models that are simple enough to compute quickly, yet detailed enough to capture how each vehicle actually moves and interacts with the other. The kinematics, dynamics, and physical coupling between the two vehicles let the controller predict how the system evolves and compute feasible, optimal trajectories.

The reference frames and notation are fixed first, followed by the kinematic and dynamic models of the USV and the UAV.

3.1 Reference Coordinate Frames and Notation

Deriving the equations of motion requires specifying the frames against which positions, orientations, and velocities are measured. Since the vessel and the drone move both with respect to the world and with respect to each other, a common frame of reference is required to describe their motion. Figure 3.1 illustrates these frames.

Inertial Reference Frame $\{\mathcal{I}\}$ This frame establishes the global coordinate system following the East-North-Up (ENU) convention, aligned with the standard ROS convention. Its origin $\mathcal{O}_I$ is fixed at an arbitrary point on the water surface within the operating area, following the usual flat-Earth convention. The axis $x^{\mathcal{I}}$ points East, $y^{\mathcal{I}}$ points North, and $z^{\mathcal{I}}$ points upward, opposite to gravity.

USV Body-Fixed Reference Frame $\{\mathcal{B}_v\}$ This frame is rigidly attached to the vessel, with origin $\mathcal{O}_v$ at its centre of gravity. This choice decouples the linear and rotational momentum equations. The forward-pointing axis $x^{\mathcal{B}_v}$ defines surge, $y^{\mathcal{B}_v}$ points to port (sway), and $z^{\mathcal{B}_v}$ points upward (heave).

UAV Body-Fixed Reference Frame $\{\mathcal{B}_a\}$ Similarly, this frame attaches to the UAV, with origin $\mathcal{O}_a$ at its centre of gravity, decoupling the translational and rotational dynamics in the same way. The longitudinal axis $x^{\mathcal{B}_a}$ points forward, the lateral axis $y^{\mathcal{B}_a}$ points to the left, and the vertical axis $z^{\mathcal{B}_a}$ points upward, aligning with the collective thrust direction.

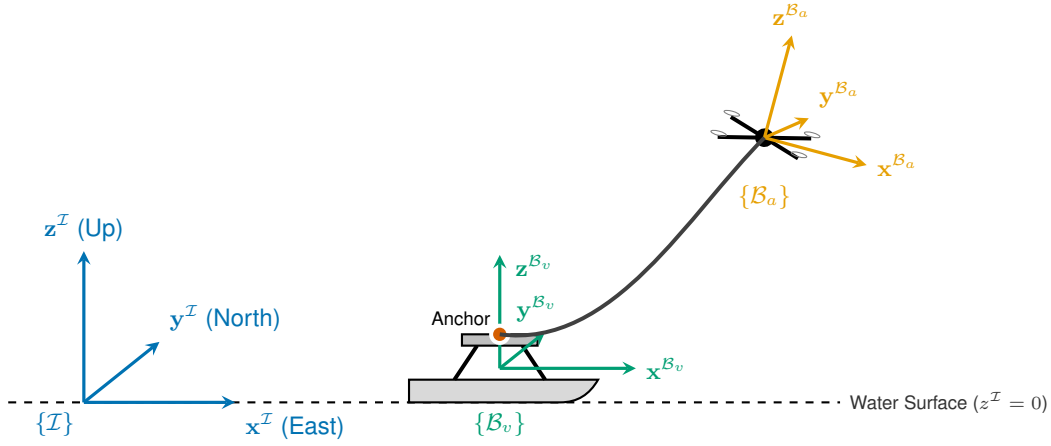


Figure 3.1: Reference coordinate frames $\{\mathcal{I}\}$, $\{\mathcal{B}_v\}$, and $\{\mathcal{B}_a\}$.

3.2 WAM-V Surface Vessel Model

The mathematical model of the WAM-V catamaran, covering its horizontal maneuvering dynamics and thruster allocation, forms a control-oriented state-space model suitable for trajectory tracking and predictive control.

Although a rigid body moving freely in the marine environment inherently has six degrees of freedom, the WAM-V is a vessel that moves on the surface, which allows the motion to be simplified to the horizontal plane. Under this assumption, the model is reduced to three degrees of freedom.

Predicting the vessel's motion, as needed for trajectory tracking and predictive control, requires knowing how its state changes over time for a given input. This relationship is captured by the nonlinear state-space model

$$\dot{\mathbf{x}}_v = f(\mathbf{x}_v, \mathbf{u}_v), \quad (3.1)$$

The state vector $\mathbf{x}_v \in \mathbb{R}^6$ is given by

$$\mathbf{x}_v = [x, y, \psi, u, v, r]^T, \quad (3.2)$$

where:

- x, y – position of the origin $\mathcal{O}_v$ of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;
- ψ – heading of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$;
- u, v – surge and sway linear velocities of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{B}_v\}$;
- r – angular velocity of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{B}_v\}$;

The control input vector $\mathbf{u}_v \in \mathbb{R}^3$ is given by

$$\mathbf{u}_v = [X, Y, N]^T, \quad (3.3)$$

where:

- X – surge force generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;
- Y – sway force generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;
- N – yaw moment generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;

The vector field $f(\cdot)$ splits into a kinematic block and a dynamic block, derived in turn in the following subsections.

3.2.1 Kinematic Model

The kinematics describe how the vessel's inertial position and heading evolve as a function of its body-fixed velocities, forming the first half of (3.1).

Let $\mathbf{p} = x \mathbf{x}^{\mathcal{I}} + y \mathbf{y}^{\mathcal{I}}$ denote the inertial position of the origin $\mathcal{O}_v$ of $\{\mathcal{B}_v\}$. Since $\mathbf{x}^{\mathcal{I}}, \mathbf{y}^{\mathcal{I}}$ are constant, differentiating yields

$$\dot{\mathbf{p}} = \dot{x} \mathbf{x}^{\mathcal{I}} + \dot{y} \mathbf{y}^{\mathcal{I}}, \quad (3.4)$$

i.e. $\dot{x}, \dot{y}$ are, by definition, the components of $\dot{\mathbf{p}}$ in the inertial frame.

The surge and sway velocities u, v are defined in the body-fixed frame, so that

$$\dot{\mathbf{p}} = u \mathbf{x}^{\mathcal{B}} + v \mathbf{y}^{\mathcal{B}}, \quad (3.5)$$

where $\mathbf{x}^{\mathcal{B}}, \mathbf{y}^{\mathcal{B}}$ denote the basis vectors of $\{\mathcal{B}_v\}$, i.e. $\mathbf{x}^{\mathcal{B}_v}, \mathbf{y}^{\mathcal{B}_v}$.

Since $\{\mathcal{B}_v\}$ is rotated relative to $\{\mathcal{I}\}$ by the heading angle ψ , the inertial unit vectors are related to the body-fixed ones by the rotation matrix $\mathbf{R}(\psi) \in SO(2)$,

$$\begin{bmatrix} \mathbf{x}^{\mathcal{I}} \\ \mathbf{y}^{\mathcal{I}} \end{bmatrix} = \mathbf{R}(\psi) \begin{bmatrix} \mathbf{x}^{\mathcal{B}} \\ \mathbf{y}^{\mathcal{B}} \end{bmatrix}, \quad (3.6)$$

following the standard planar rotation matrix,

$$\mathbf{R}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}, \quad (3.7)$$

which is equivalent to

$$\mathbf{x}^{\mathcal{B}} = \cos \psi \mathbf{x}^{\mathcal{I}} + \sin \psi \mathbf{y}^{\mathcal{I}}, \quad \mathbf{y}^{\mathcal{B}} = -\sin \psi \mathbf{x}^{\mathcal{I}} + \cos \psi \mathbf{y}^{\mathcal{I}}. \quad (3.8)$$

Substituting (3.8) into (3.5) and grouping terms gives

$$\dot{\mathbf{p}} = \underbrace{(u \cos \psi - v \sin \psi)}_{\dot{x}} \mathbf{x}^{\mathcal{I}} + \underbrace{(u \sin \psi + v \cos \psi)}_{\dot{y}} \mathbf{y}^{\mathcal{I}}. \quad (3.9)$$

Comparing with (3.4) then directly gives

$$\dot{x} = u \cos \psi - v \sin \psi, \quad \dot{y} = u \sin \psi + v \cos \psi, \quad (3.10)$$

which is precisely

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \mathbf{R}(\psi) \begin{bmatrix} u \\ v \end{bmatrix}. \quad (3.11)$$

The heading rate follows directly from the definition of the angular velocity r :

$$\dot{\psi} = r. \quad (3.12)$$

3.2.2 Dynamic Model

The dynamics describe how the vessel's body-fixed velocities evolve as a function of the applied forces and moments, forming the second half of (3.1).

Unlike $\mathbf{x}^I, \mathbf{y}^I$, the body-fixed unit vectors $\mathbf{x}^B, \mathbf{y}^B$ rotate with the vehicle. Differentiating (3.8) and using $\dot{\psi} = r$ gives

$$\dot{\mathbf{x}}^B = r \mathbf{y}^B, \quad \dot{\mathbf{y}}^B = -r \mathbf{x}^B. \quad (3.13)$$

The inertial velocity, expressed in the body-fixed frame, is $\dot{\mathbf{p}} = u \mathbf{x}^B + v \mathbf{y}^B$. Differentiating once more and substituting (3.13) yields the inertial acceleration

$$\ddot{\mathbf{p}} = (\dot{u} - vr) \mathbf{x}^B + (\dot{v} + ur) \mathbf{y}^B, \quad (3.14)$$

where the cross terms $-vr$ and ur are Coriolis/centripetal terms arising because $\{\mathcal{B}_v\}$ is a rotating, non-inertial frame.

Let F_x, F_y denote the resultant surge and sway forces applied at $\mathcal{O}_v$, so that $\mathbf{F} = F_x \mathbf{x}^B + F_y \mathbf{y}^B$. By Newton's second law, $\mathbf{F} = m \ddot{\mathbf{p}}$, and comparing with (3.14),

$$F_x = m(\dot{u} - vr), \quad F_y = m(\dot{v} + ur). \quad (3.15)$$

Since the vessel's motion is planar, Euler's rotational equation about $\mathbf{z}^B$ takes the form

$$M_z = I_z \dot{r}, \quad (3.16)$$

where M_z is the resultant yaw moment about $\mathcal{O}_v$ and I_z the yaw moment of inertia.

The resultant force and moment F_x, F_y, M_z combine two contributions: the thrust generated by the propulsion system, X, Y, N , and the hydrodynamic damping opposing the vessel's motion. This damping follows the standard linear-plus-quadratic model for surface vessels [22]. The quadratic surge term is further split into forward and backward coefficients, to capture the asymmetric drag between the two

directions:

$$F_x = X - (X_u + X_{uu}^\pm |u|)u, \quad X_{uu}^\pm = \begin{cases} X_{uu}^+, & u \geq 0 \\ X_{uu}^-, & u < 0 \end{cases} \quad (3.17)$$

$$F_y = Y - (Y_v + Y_{vv} |v|)v, \quad M_z = N - (N_r + N_{rr} |r|)r. \quad (3.18)$$

Substituting (3.17)–(3.18) into (3.15) and (3.16), and solving for the accelerations, gives the dynamic block of (3.1):

$$\dot{u} = \frac{1}{m} [X - (X_u + X_{uu}^\pm |u|)u] + vr, \quad (3.19)$$

$$\dot{v} = \frac{1}{m} [Y - (Y_v + Y_{vv} |v|)v] - ur, \quad (3.20)$$

$$\dot{r} = \frac{1}{I_z} [N - (N_r + N_{rr} |r|)r]. \quad (3.21)$$

3.2.3 Thruster Allocation

The thruster allocation maps the physical actuator commands, the left and right thrust magnitudes T_L, T_R and azimuth angles α_L, α_R , to the resultant force and moment X, Y, N required by the dynamics of Section 3.2.2, according to the geometry shown in Figure 3.2.

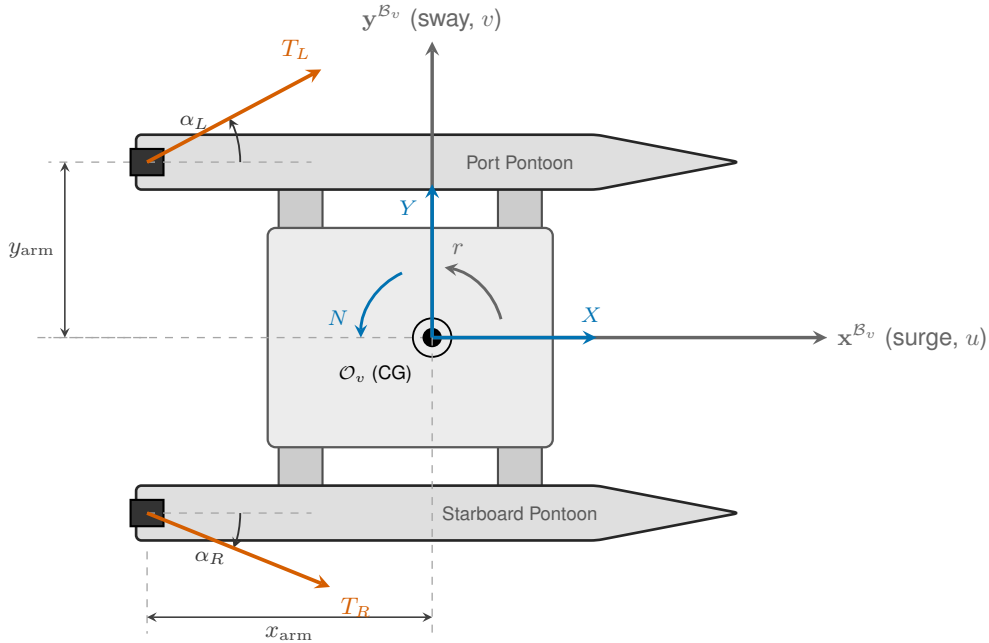


Figure 3.2: Planar USV schematic and thruster allocation geometry.

Each azimuth angle is measured relative to $\mathbf{x}^B$, so that each thruster produces a planar force

$$\mathbf{f}_L = T_L \cos \alpha_L \mathbf{x}^B + T_L \sin \alpha_L \mathbf{y}^B, \quad \mathbf{f}_R = T_R \cos \alpha_R \mathbf{x}^B + T_R \sin \alpha_R \mathbf{y}^B. \quad (3.22)$$

Summing the components of both thrusters along $\mathbf{x}^B$ and $\mathbf{y}^B$ gives the resultant force applied at $\mathcal{O}_v$:

$$X = T_L \cos \alpha_L + T_R \cos \alpha_R, \quad Y = T_L \sin \alpha_L + T_R \sin \alpha_R. \quad (3.23)$$

Both thrusters are mounted behind $\mathcal{O}_v$, at the same longitudinal distance x_{arm} , and symmetrically on either side of the centerline, at a lateral distance y_{arm} . Expressed in $\{\mathcal{B}_v\}$, the left (port) thruster is located at $(-x_{\text{arm}}, y_{\text{arm}})$ and the right (starboard) thruster at $(-x_{\text{arm}}, -y_{\text{arm}})$.

The yaw moment produced by a planar force (f_x, f_y) applied at (r_x, r_y) is given by the two-dimensional cross product $r_x f_y - r_y f_x$. Applying this to each thruster and summing gives the resultant yaw moment about $\mathcal{O}_v$:

$$N = -x_{\text{arm}}(T_L \sin \alpha_L + T_R \sin \alpha_R) - y_{\text{arm}}(T_L \cos \alpha_L - T_R \cos \alpha_R). \quad (3.24)$$

Equations (3.23) and (3.24) complete the mapping from the four physical actuator commands $T_L, T_R, \alpha_L, \alpha_R$ to the control input $\mathbf{u}_v = [X, Y, N]^T$ used by the dynamic model.

3.3 x500 Quadrotor Model

The mathematical model of the x500 quadrotor, covering its translational dynamics in three-dimensional space, forms a control-oriented state-space model suitable for trajectory tracking and predictive control. The vehicle's attitude is stabilized by an inner-loop attitude controller, to which the attitude used in the translational dynamics is commanded.

Predicting the vehicle's motion, as needed for trajectory tracking and predictive control, requires knowing how its state changes over time for a given input. This relationship is captured by the nonlinear state-space model

$$\dot{\mathbf{x}}_a = f(\mathbf{x}_a, \mathbf{u}_a), \quad (3.25)$$

The state vector $\mathbf{x}_a \in \mathbb{R}^6$ is given by

$$\mathbf{x}_a = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^T, \quad (3.26)$$

where:

- x, y, z – position of the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;
- $\dot{x}, \dot{y}, \dot{z}$ – linear velocity of $\mathcal{O}_a$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;

The control input vector $\mathbf{u}_a \in \mathbb{R}^4$ is given by

$$\mathbf{u}_a = [T, \phi, \theta, \psi]^T, \quad (3.27)$$

where:

- T – total thrust force, applied to $\{\mathcal{B}_a\}$, expressed in $\{\mathcal{B}_a\}$;
- ϕ, θ, ψ – roll, pitch, and yaw angles of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$;

The vector field $f(\cdot)$ splits into a kinematic block, relating the inertial position and velocity, and a dynamic block, mapping the thrust, attitude, and tether force to the inertial translational acceleration.

3.3.1 Kinematic Model

The kinematics describe how the vehicle's inertial position evolves as a function of its inertial velocity, forming the first half of (3.25).

Let $\mathbf{p} = x \mathbf{x}^{\mathcal{I}} + y \mathbf{y}^{\mathcal{I}} + z \mathbf{z}^{\mathcal{I}}$ denote the inertial position of the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$. Since $\mathbf{x}^{\mathcal{I}}, \mathbf{y}^{\mathcal{I}}, \mathbf{z}^{\mathcal{I}}$ are constant, differentiating yields

$$\dot{\mathbf{p}} = \dot{x} \mathbf{x}^{\mathcal{I}} + \dot{y} \mathbf{y}^{\mathcal{I}} + \dot{z} \mathbf{z}^{\mathcal{I}}, \quad (3.28)$$

i.e. $\dot{x}, \dot{y}, \dot{z}$ are, by definition, the components of $\dot{\mathbf{p}}$ in the inertial frame, which are precisely the linear velocity components in $\mathbf{x}_a$. The kinematic block of (3.25) is therefore the trivial identity

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}. \quad (3.29)$$

This is in contrast to a surface vessel such as the WAM-V (Section 3.2), where the linear velocities are instead expressed in $\{\mathcal{B}_v\}$, requiring an additional change of basis via $\mathbf{R}(\psi)$ to relate them to $\dot{\mathbf{p}}$. Here, this rotation is required instead to express the thrust force, applied along $\mathbf{z}^{\mathcal{B}_a}$, in $\{\mathcal{I}\}$. It is therefore deferred to the dynamic block, derived next.

3.3.2 Dynamic Model

The dynamics describe how the vehicle's inertial velocity evolves as a function of the applied thrust, attitude, and tether force, forming the second half of (3.25).

The thrust generated by the propulsion system acts along the vertical axis of $\{\mathcal{B}_a\}$, so that the resultant thrust force is

$$\mathbf{F}_{\text{thr}} = T \mathbf{z}^{\mathcal{B}}, \quad (3.30)$$

where $\mathbf{z}^{\mathcal{B}}$ denotes the vertical basis vector of $\{\mathcal{B}_a\}$, i.e. $\mathbf{z}^{\mathcal{B}_a}$.

Since (3.29) relates the inertial position and velocity directly, without an intermediate change of basis, Newton's second law is more conveniently applied directly in $\{\mathcal{I}\}$, requiring $\mathbf{F}_{\text{thr}}$ to instead be expressed in the inertial frame.

Since $\{\mathcal{B}_a\}$ is rotated relative to $\{\mathcal{I}\}$ by the roll, pitch, and yaw angles ϕ, θ, ψ , the inertial unit vectors are related to the body-fixed ones by the rotation matrix $\mathbf{R}(\phi, \theta, \psi) \in SO(3)$,

$$\begin{bmatrix} \mathbf{x}^{\mathcal{I}} \\ \mathbf{y}^{\mathcal{I}} \\ \mathbf{z}^{\mathcal{I}} \end{bmatrix} = \mathbf{R}(\phi, \theta, \psi) \begin{bmatrix} \mathbf{x}^{\mathcal{B}} \\ \mathbf{y}^{\mathcal{B}} \\ \mathbf{z}^{\mathcal{B}} \end{bmatrix}, \quad (3.31)$$

following the ZYX (yaw-pitch-roll) Euler angle convention,

$$\mathbf{R}(\phi, \theta, \psi) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \quad (3.32)$$

where $s_\bullet = \sin(\bullet)$ and $c_\bullet = \cos(\bullet)$.

Projecting $\mathbf{F}_{\text{thr}}$ onto $\{\mathcal{I}\}$ then gives

$$\mathbf{F}_{\text{thr}}^{\mathcal{I}} = \mathbf{R}(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}. \quad (3.33)$$

Figure 3.3 illustrates this projection in the longitudinal plane, showing how the body tilt θ splits the thrust force T into vertical and horizontal inertial components.

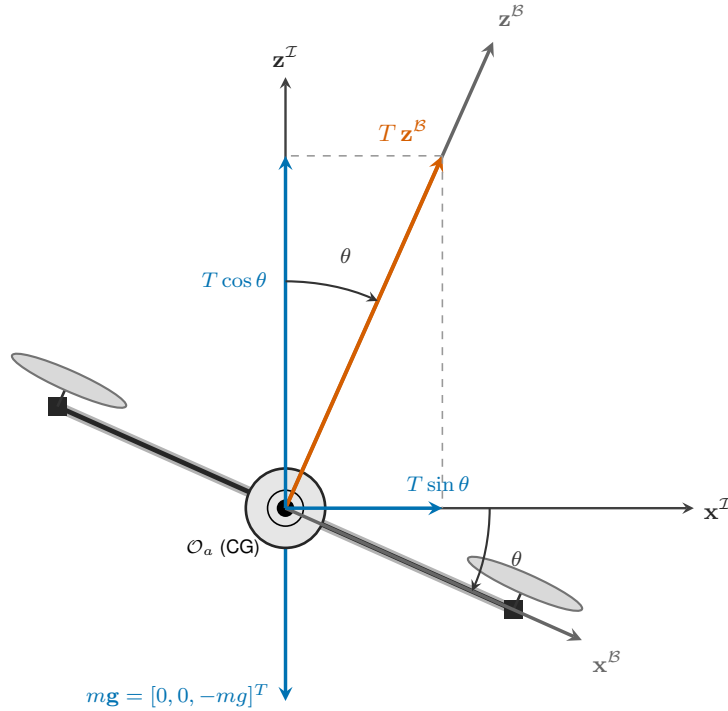


Figure 3.3: Quadrotor thrust vectoring in the longitudinal plane.

Gravity, unlike the thrust, is already naturally expressed in $\{\mathcal{I}\}$:

$$\mathbf{F}_{\text{grav}}^{\mathcal{I}} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}. \quad (3.34)$$

In addition to thrust and gravity, the vehicle is subject to a force transmitted through a tether attached at the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$, denoted $\mathbf{F}_{\text{tether}}^{\mathcal{I}}$ and derived separately in Section 3.3.3.

Since $\{\mathcal{I}\}$ is non-rotating, Newton's second law applies directly to the inertial acceleration, without the transport terms that arise when applying it in a rotating frame, such as $\{\mathcal{B}_v\}$ in the WAM-V model (Section 3.2.2):

$$m \ddot{\mathbf{p}} = \mathbf{F}_{\text{thr}}^{\mathcal{I}} + \mathbf{F}_{\text{grav}}^{\mathcal{I}} + \mathbf{F}_{\text{tether}}^{\mathcal{I}}. \quad (3.35)$$

Substituting (3.33) and (3.34) into (3.35), and solving for the acceleration, gives the dynamic block of (3.25):

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \frac{T}{m} \mathbf{R}(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} + \frac{1}{m} \mathbf{F}_{\text{tether}}^{\mathcal{I}}. \quad (3.36)$$

3.3.3 Tether Force Model

In addition to thrust and gravity, the vehicle is subject to a force transmitted through a tether connecting the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$ to a fixed attachment point on the WAM-V, whose position $\mathbf{p}_v$ is treated here only as a known geometric anchor, without modeling the (negligible) reaction force exerted on the boat.

The relevant geometric quantity is the straight-line vector between the two attachment points,

$$\mathbf{r} = \mathbf{p}_v - \mathbf{p}, \quad d = \|\mathbf{r}\|, \quad \hat{\mathbf{d}}_0 = \frac{\mathbf{r}}{d}, \quad (3.37)$$

where d is the straight-line distance between the vehicle and the boat, and $\hat{\mathbf{d}}_0$ is the direction the tether would have if taut.

Denoting by $L \geq d$ the tether length at a given instant, its slack is conveniently quantified by the dimensionless ratio

$$\varepsilon = \frac{L - d}{L}, \quad (3.38)$$

with $\varepsilon = 0$ corresponding to a taut tether, and ε increasing as more slack is introduced.

Under its own weight, a tether hanging between two fixed points takes the shape of a catenary, contained in the vertical plane defined by the two attachment points. In this plane, the exact catenary equations relate ε to the angle γ by which the tension direction at $\mathcal{O}_a$ deviates from $\hat{\mathbf{d}}_0$ towards the vertical, and to the tension magnitude T_{tether} at that point.

These relations, denoted $\gamma_{\text{exact}}(\varepsilon)$ and $T_{\text{tether}}^{\text{exact}}(\varepsilon)$, do not admit a closed-form solution. Evaluating them exactly therefore requires an iterative numerical solution at every time step.

The tether control scheme considered in this work maintains ε close to a constant operating point ε_0 . This makes it possible to avoid repeatedly solving these equations as the vehicle moves, by approximating them locally about ε_0 .

This motivates approximating $\gamma_{\text{exact}}(\cdot)$ and $T_{\text{tether}}^{\text{exact}}(\cdot)$ by their first-order Taylor expansions about ε_0 ,

$$\gamma(\varepsilon) = \gamma(\varepsilon_0) + \gamma'(\varepsilon_0) (\varepsilon - \varepsilon_0), \quad T_{\text{tether}}(\varepsilon) = T_{\text{tether}}(\varepsilon_0) + T_{\text{tether}}'(\varepsilon_0) (\varepsilon - \varepsilon_0), \quad (3.39)$$

which replace the exact relations with closed-form expressions whose coefficients are computed offline, once, at ε_0 .

With $\gamma(\varepsilon)$ given by the approximation in (3.39), the tension direction $\hat{\mathbf{d}}(\varepsilon)$ can now be constructed. This requires first introducing the direction $\hat{\mathbf{e}}_h$, orthogonal to $\mathbf{z}^{\mathcal{I}}$ within the catenary plane,

$$\hat{\mathbf{e}}_h = \frac{\hat{\mathbf{d}}_0 - (\hat{\mathbf{d}}_0 \cdot \mathbf{z}^{\mathcal{I}}) \mathbf{z}^{\mathcal{I}}}{\|\hat{\mathbf{d}}_0 - (\hat{\mathbf{d}}_0 \cdot \mathbf{z}^{\mathcal{I}}) \mathbf{z}^{\mathcal{I}}\|}, \quad (3.40)$$

and the depression angle α of $\hat{\mathbf{d}}_0$ below $\hat{\mathbf{e}}_h$, which follows from

$$\hat{\mathbf{d}}_0 = \cos \alpha \hat{\mathbf{e}}_h - \sin \alpha \mathbf{z}^{\mathcal{I}}. \quad (3.41)$$

Figure 3.4 illustrates this construction within the catenary plane, showing the local horizontal $\hat{\mathbf{e}}_h$, the depression angle α , and the resulting tension direction $\hat{\mathbf{d}}(\varepsilon)$, tilted further from $\hat{\mathbf{d}}_0$ by the angle γ .

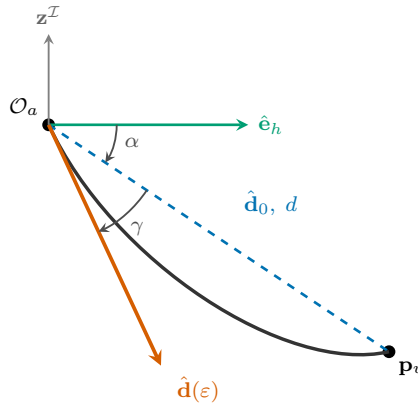


Figure 3.4: Tether catenary plane.

The tension direction at $\mathcal{O}_a$ then follows by tilting $\hat{\mathbf{d}}_0$ further towards the vertical by the angle $\gamma(\varepsilon)$, within the same plane,

$$\hat{\mathbf{d}}(\varepsilon) = \cos(\alpha + \gamma(\varepsilon)) \hat{\mathbf{e}}_h - \sin(\alpha + \gamma(\varepsilon)) \mathbf{z}^{\mathcal{I}}, \quad (3.42)$$

and the resulting tether force, applied at $\mathcal{O}_a$ and expressed in $\{\mathcal{I}\}$, is

$$\mathbf{F}_{\text{tether}}^{\mathcal{I}} = T_{\text{tether}}(\varepsilon) \hat{\mathbf{d}}(\varepsilon), \quad (3.43)$$

with ε evaluated at each instant from (3.37)–(3.38). Equation (3.43) is the term $\mathbf{F}_{\text{tether}}^{\mathcal{I}}$ introduced in (3.35).

Chapter 4

System Control

Having established, in Chapter 3, the kinematic and dynamic models governing the USV and UAV, the natural next step is to determine how these models can be exploited to compute the control inputs that drive each vehicle along a desired trajectory. Because the two vehicles are physically coupled through the tether, their motion cannot be planned independently without accounting for this interaction, motivating a cooperative control strategy.

The overall control approach is therefore established first, followed by the design of the controllers for each vehicle.

4.1 Cooperative Control Design

Before formulating either controller, the strategies available are weighed against the system's dynamics and its objectives, so that the methods chosen suits both vehicles.

4.1.1 Control Method

Keeping the tether within safe operating conditions depends on how the distance between the two vehicles evolves over time. A controller that only reacts to the current state can correct a violation only after it happens. Anticipating where the system is headed allows corrective action before reaching the safety limits.

A predictive controller does this by repeatedly solving, at each control instant, an optimization problem over a finite look-ahead window. Because the optimization runs over the predicted trajectory rather than the current state alone, requirements such as keeping the tether within safe conditions can be stated as explicit constraints in the problem itself, guaranteeing safety limits by construction over the entire prediction horizon.

The predictions rely on the model of the system dynamics, so their accuracy is tied to how well that model represents the real behaviour. The USV and UAV models derived in Chapter 3 include nonlinear terms. The predictive controller uses these nonlinear models directly, making *Nonlinear Model Predictive Control* (NMPC) the method used for both vehicles.

4.1.2 Cooperative Architecture

The USV and the UAV operate on different timescales. The USV moves in the horizontal plane with slow, inertia-dominated motion. The UAV's dynamics evolve much faster and demand a finer sampling time. A single NMPC built around the combined state of both vehicles would force one prediction horizon and one sampling time onto both, compromising one vehicle's timescale to accommodate the other.

Combining both vehicles into one NMPC also enlarges the state, input, and constraint sets a single solver must handle, raising the computational cost of a problem that is already time-critical. Keeping the two problems separate matches each optimization's size to its own vehicle and keeps the real-time computation manageable.

Each vehicle is therefore controlled by its own NMPC. This separation isolates the two controllers from each other's failures, if one solver fails to find a feasible solution, the other vehicle's NMPC keeps operating on its own well-posed problem.

The tether's safety limits depend on the relative position between the USV and the UAV, so each vehicle's NMPC accounts for the other vehicle's position. At every control instant, each controller can receive the other vehicle's predicted position over the full prediction horizon and use it to compute the predicted tether state at every step of its own horizon.

Without this exchange, a vehicle would have to treat the other as stationary or extrapolate its motion from the last known state. Either assumption misrepresents the tether coupling and can drive the system into a constraint violation.

Figure 4.1 shows this architecture, two independent NMPC loops, each driving its own vehicle, exchanging predicted horizons to enforce the tether constraint.

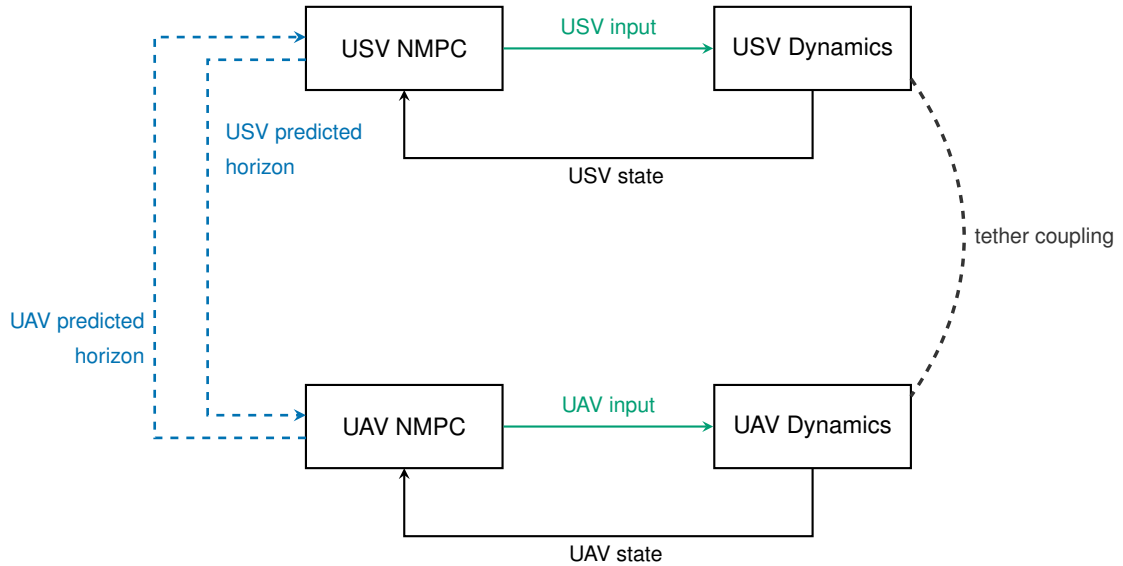


Figure 4.1: Cooperative control architecture diagram.

4.2 Nonlinear Model Predictive Control

The vehicle models of Chapter 3 are continuous-time systems of the form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad (4.1)$$

where $\mathbf{x} \in \mathbb{R}^{n_x}$ is the state vector and $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control input vector. Since the controller acts at discrete sampling instants $t_k = kT_s$, with the input held constant over each interval $[t_k, t_{k+1})$, the model is converted into the discrete-time system

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k, \mathbf{u}_k) = \mathbf{x}_k + \int_{t_k}^{t_{k+1}} \mathbf{f}(\mathbf{x}(\tau), \mathbf{u}_k) d\tau, \quad (4.2)$$

obtained by numerical integration over T_s . NMPC computes, at every instant k , the input that steers this system toward a reference over a finite look-ahead window of N steps.

At instant k , denote by $\hat{\mathbf{x}}_{k+i|k}$ and $\hat{\mathbf{u}}_{k+i|k}$ the state and input predicted i steps ahead, for $i = 0, \dots, N$. The hat marks a predicted quantity rather than the true system state. The predicted trajectory starts from the current measured or estimated state,

$$\hat{\mathbf{x}}_{k|k} = \mathbf{x}_k, \quad (4.3)$$

and evolves according to the system model,

$$\hat{\mathbf{x}}_{k+i+1|k} = \mathbf{F}(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}), \quad i = 0, \dots, N-1. \quad (4.4)$$

Each predicted step is then scored against a reference trajectory. A stage cost $\ell(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k})$ penalizes the tracking error and the control effort at every step of the horizon, and a terminal cost $\ell_f(\hat{\mathbf{x}}_{k+N|k})$ penalizes the tracking error at its final step. Summing both over the horizon gives the total predicted cost,

$$J = \sum_{i=0}^{N-1} \ell(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) + \ell_f(\hat{\mathbf{x}}_{k+N|k}). \quad (4.5)$$

Minimizing J alone ignores the physical limits of the system, so the prediction is also constrained. Direct bounds apply to the state and the input at every step of the horizon,

$$\hat{\mathbf{x}}_{k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \quad (4.6)$$

$$\hat{\mathbf{u}}_{k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \quad (4.7)$$

where $\mathcal{X}$ and $\mathcal{U}$ collect the admissible states and inputs. Additional requirements that combine state and input, or that cannot be expressed as simple bounds, are stated as a general path constraint,

$$\mathbf{g}(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq \mathbf{0}, \quad i = 0, \dots, N-1. \quad (4.8)$$

Collecting the cost, the prediction model, and the constraints defines the optimal control problem

solved at every sampling instant k :

$$\begin{aligned}
& \min_{\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}} J \\
& \text{s.t. } \hat{\mathbf{x}}_{k|k} = \mathbf{x}_k, \\
& \quad \hat{\mathbf{x}}_{k+i+1|k} = \mathbf{F}(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}), \quad i = 0, \dots, N-1, \\
& \quad \hat{\mathbf{x}}_{k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \\
& \quad \hat{\mathbf{u}}_{k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \\
& \quad \mathbf{g}(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq \mathbf{0}, \quad i = 0, \dots, N-1.
\end{aligned} \tag{4.9}$$

Solving this problem returns a full predicted input sequence, $\hat{\mathbf{u}}_{k|k}, \dots, \hat{\mathbf{u}}_{k+N-1|k}$, but only its first element, $\hat{\mathbf{u}}_{k|k}$, is applied to the system. At the next instant, $k+1$, the state is measured again and the problem is solved from scratch, now anchored at $\hat{\mathbf{x}}_{k+1|k+1}$. This is the receding horizon principle: the horizon slides forward with time, and every applied input comes from a fresh optimization over the current prediction, not from a plan fixed in advance.

4.3 USV NMPC

The USV NMPC computes, at every sampling instant, the thruster commands that best track a reference trajectory over a finite prediction horizon, subject to the vessel's dynamics and the physical limits of its actuators. It predicts the vessel's motion with the model derived in Chapter 3, and solves a constrained optimization problem at each step to obtain the commands applied to the thrusters.

Predictive Model

The input $\mathbf{u}_v = [X, Y, N]^T$ of the vessel model is the resultant thrust force and moment, obtained from the physical actuator commands $T_L, T_R, \alpha_L, \alpha_R$ through the allocation

$$X = T_L \cos \alpha_L + T_R \cos \alpha_R, \quad Y = T_L \sin \alpha_L + T_R \sin \alpha_R, \tag{4.10}$$

$$N = -x_{arm}(T_L \sin \alpha_L + T_R \sin \alpha_R) - y_{arm}(T_L \cos \alpha_L - T_R \cos \alpha_R). \tag{4.11}$$

Since the controller's ultimate goal is always to command $T_L, T_R, \alpha_L, \alpha_R$, the most direct approach is to let the NMPC choose them itself, rather than choosing X, Y, N and allocating separately. This also guarantees that every value chosen by the NMPC is attainable, since the bounds are imposed directly on the physical actuator commands, rather than on an approximation of their combined effect. This amounts to redefining the input vector as $\mathbf{u}_v = [T_L, T_R, \alpha_L, \alpha_R]^T$.

However the thrusters can only change thrust and azimuth at a limited rate,

$$|\dot{T}_L| \leq \dot{T}_{L,max}, \quad |\dot{T}_R| \leq \dot{T}_{R,max}, \quad |\dot{\alpha}_L| \leq \dot{\alpha}_{L,max}, \quad |\dot{\alpha}_R| \leq \dot{\alpha}_{R,max}. \tag{4.12}$$

To impose (4.12) directly, $T_L, T_R, \alpha_L, \alpha_R$ can be treated as states rather than as the input,

$$\mathbf{x}_v = [x, y, \psi, u, v, r, T_L, T_R, \alpha_L, \alpha_R]^T \in \mathbb{R}^{10}. \quad (4.13)$$

Their time derivative can then be used as the input, allowing the rate constraint to be applied directly to it,

$$\mathbf{u}_v = [\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^T \in \mathbb{R}^4, \quad (4.14)$$

so that (4.12) becomes a direct bound on $\mathbf{u}_v$.

With $\mathbf{x}_v$ and $\mathbf{u}_v$ so defined, the dynamics $\dot{\mathbf{x}}_v = f(\mathbf{x}_v, \mathbf{u}_v)$ retain the same form in their first six components,

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\psi} \\ \dot{u} \\ \dot{v} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} u \cos \psi - v \sin \psi \\ u \sin \psi + v \cos \psi \\ r \\ \frac{1}{m} [X - (X_u + X_{uu}|u|)u] + vr \\ \frac{1}{m} [Y - (Y_v + Y_{vv}|v|)v] - ur \\ \frac{1}{I_z} [N - (N_r + N_{rr}|r|)r] \end{bmatrix}, \quad \begin{bmatrix} \dot{T}_L \\ \dot{T}_R \\ \dot{\alpha}_L \\ \dot{\alpha}_R \end{bmatrix} = \mathbf{u}_v, \quad (4.15)$$

where X, Y, N are given by (4.10)–(4.11) in terms of the thruster state, and $\mathbf{u}_v$ appears directly as the derivative of that same state.

Discretization over a sampling interval T_s , following Section 4.2, gives the one-step map F such that $\hat{\mathbf{x}}_{k+i+1|k} = F(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k})$ approximates the solution of $\dot{\mathbf{x}}_v = f(\mathbf{x}_v, \mathbf{u}_v)$ over one sampling interval.

Cost and Constraints

Let $\mathbf{p} = [x, y]^T$ and $\boldsymbol{\nu} = [u, v, r]^T$ denote the position and body-fixed velocities, and ψ the heading, with $\mathbf{p}_{ref}$, ψ_{ref} , and $\boldsymbol{\nu}_{ref}$ the corresponding references.

Tracking the reference position is penalized as

$$\|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2, \quad (4.16)$$

where $\|\mathbf{v}\|_Q^2 := \mathbf{v}^T Q \mathbf{v}$ for a positive semi-definite weighting matrix Q , which sets the relative importance of each component of $\mathbf{v}$ in the cost.

Both ψ and ψ_{ref} are angles, and are therefore each confined to an interval of width 2π , such as $(-\pi, \pi]$. When they lie near opposite ends of this interval, their raw difference $\psi - \psi_{ref}$ approaches $\pm 2\pi$, even though the two headings are then almost identical. The heading is therefore tracked as

$$q_\psi (1 - \cos(\psi - \psi_{ref})). \quad (4.17)$$

As $\cos(\cdot)$ has period exactly 2π , $\cos(\psi - \psi_{ref})$ evaluates to exactly the same value whether $\psi - \psi_{ref}$ is close to 0 or close to $\pm 2\pi$, so (4.17) correctly assigns a similarly small cost in both cases.

Tracking the reference velocity is penalized in the same way as the position,

$$\|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|_{Q_\nu}^2. \quad (4.18)$$

Unlike the previous three terms, the next one does not compare against any reference. It penalizes $T_L, T_R, \alpha_L, \alpha_R$ directly,

$$\|[T_L, T_R, \alpha_L, \alpha_R]^T\|_Q^2, \quad (4.19)$$

discouraging thrust and azimuth commands larger than necessary to track the reference.

The last term penalizes $\mathbf{u}_v$, the rate of change of the thruster commands,

$$\|[\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^T\|_R^2 = \|\mathbf{u}_v\|_R^2, \quad (4.20)$$

discouraging abrupt changes in thrust and azimuth between consecutive instants.

Summing (4.16)–(4.20) gives the stage cost

$$\ell(\mathbf{x}_v, \mathbf{u}_v) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + q_\psi(1 - \cos(\psi - \psi_{ref})) + \|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|_{Q_\nu}^2 + \|[T_L, T_R, \alpha_L, \alpha_R]^T\|_Q^2 + \|\mathbf{u}_v\|_R^2. \quad (4.21)$$

The terminal cost ℓ_f penalizes the tracking error of the last predicted state, preventing the finite horizon from letting that error grow unchecked near its end. It keeps the tracking terms of (4.21) and drops those on the thruster commands and their rates, since no input is chosen at the final node of the horizon,

$$\ell_f(\mathbf{x}_v) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + q_\psi(1 - \cos(\psi - \psi_{ref})) + \|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|_{Q_\nu}^2. \quad (4.22)$$

Each thruster is limited both in the values $T_L, T_R, \alpha_L, \alpha_R$ can take and in how fast they can change. Since these commands are now components of $\mathbf{x}_v$, the first limit becomes a bound on the state,

$$\mathcal{X} : T_{L,min} \leq T_L \leq T_{L,max}, \quad T_{R,min} \leq T_R \leq T_{R,max}, \quad \alpha_{L,min} \leq \alpha_L \leq \alpha_{L,max}, \quad \alpha_{R,min} \leq \alpha_R \leq \alpha_{R,max}, \quad (4.23)$$

and since their rate is now $\mathbf{u}_v$, the second becomes a bound on the input,

$$\mathcal{U} : |\dot{T}_L| \leq \dot{T}_{L,max}, \quad |\dot{T}_R| \leq \dot{T}_{R,max}, \quad |\dot{\alpha}_L| \leq \dot{\alpha}_{L,max}, \quad |\dot{\alpha}_R| \leq \dot{\alpha}_{R,max}. \quad (4.24)$$

Optimal Control Problem

At every sampling instant k , the kinematic and dynamic components of the initial state come from the vessel's current estimate, while the thruster components use the last command actually applied,

$$\hat{\mathbf{x}}_{v,k|k} = [x_k, y_k, \psi_k, u_k, v_k, r_k, T_{L,k-1}, T_{R,k-1}, \alpha_{L,k-1}, \alpha_{R,k-1}]^T. \quad (4.25)$$

This ensures that the first step of the horizon already departs from the actuators' true state, and therefore respects the rate limit from the very first input chosen.

Collecting the cost, the model, and the constraints gives the optimal control problem the USV solves at every sampling instant,

$$\begin{aligned}
& \min_{\hat{\mathbf{x}}_v, \hat{\mathbf{u}}_v} \sum_{i=0}^{N-1} \ell(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}) + \ell_f(\hat{\mathbf{x}}_{v,k+N|k}) \\
& \text{s.t. } \hat{\mathbf{x}}_{v,k|k} = [x_k, y_k, \psi_k, u_k, v_k, r_k, T_{L,k-1}, T_{R,k-1}, \alpha_{L,k-1}, \alpha_{R,k-1}]^T, \\
& \quad \hat{\mathbf{x}}_{v,k+i+1|k} = F(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}), \quad i = 0, \dots, N-1, \\
& \quad \hat{\mathbf{x}}_{v,k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \\
& \quad \hat{\mathbf{u}}_{v,k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1,
\end{aligned} \tag{4.26}$$

where F is the one-step map obtained by discretizing $f(\cdot)$. The vessel receives the first input $\hat{\mathbf{u}}_{v,k|k}$ of the solution; applying it over one sampling interval yields $T_{L,k}, T_{R,k}, \alpha_{L,k}, \alpha_{R,k}$, which become the thruster components of the initial condition at the next instant.

4.4 UAV NMPC

The UAV NMPC computes, at every sampling instant, the thrust and attitude commands that best track a reference trajectory relative to the vessel over a finite prediction horizon, subject to the vehicle's dynamics, its actuation limits, and the tether workspace. It predicts the vehicle's motion with the model derived in Chapter 3, and solves a constrained optimization problem at each step to obtain the commands applied to the vehicle.

Predictive Model

The input $\mathbf{u}_a = [T, \phi, \theta, \psi]^T$ of the vehicle model commands the thrust and the attitude directly, assuming each is reached instantaneously. The vehicle's attitude, however, is stabilized by an inner-loop controller of finite bandwidth, so the roll and pitch entering the translational dynamics lag behind any commanded value. Modeling this response as a first-order lag with time constant τ gives

$$\dot{\phi} = \frac{\phi_{cmd} - \phi}{\tau}, \quad \dot{\theta} = \frac{\theta_{cmd} - \theta}{\tau}, \tag{4.27}$$

where ϕ_{cmd}, θ_{cmd} are the commanded roll and pitch. This makes ϕ, θ variables of the predictive model, reached gradually rather than instantaneously, with ϕ_{cmd}, θ_{cmd} taking the place of ϕ, θ as the input.

The attitude commands ϕ_{cmd}, θ_{cmd} can, in turn, only change at a limited rate. To impose this rate directly, ϕ_{cmd}, θ_{cmd} can be treated as states rather than as the input,

$$\mathbf{x}_a = [x, y, z, \dot{x}, \dot{y}, \dot{z}, \phi, \theta, \phi_{cmd}, \theta_{cmd}]^T \in \mathbb{R}^{10}. \tag{4.28}$$

Their time derivative can then be used as the input, allowing the rate constraint to be applied directly to

it,

$$\mathbf{u}_a = [\dot{\phi}_{cmd}, \dot{\theta}_{cmd}, a_T]^T \in \mathbb{R}^3, \quad (4.29)$$

where $a_T = T/m$ is the thrust acceleration.

With $\mathbf{x}_a$ and $\mathbf{u}_a$ so defined, the dynamics $\dot{\mathbf{x}}_a = f(\mathbf{x}_a, \mathbf{u}_a)$ are

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \ddot{x} \\ \ddot{y} \\ \ddot{z} \\ \dot{\phi} \\ \dot{\theta} \\ \dot{\phi}_{cmd} \\ \dot{\theta}_{cmd} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ a_T (\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \\ a_T (\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \\ a_T \cos \theta \cos \phi - g \\ (\phi_{cmd} - \phi)/\tau \\ (\theta_{cmd} - \theta)/\tau \\ \dot{\phi}_{cmd} \\ \dot{\theta}_{cmd} \end{bmatrix}, \quad (4.30)$$

where ψ enters as an external parameter rather than as a variable chosen by the NMPC, and $\dot{\phi}_{cmd}, \dot{\theta}_{cmd}$ appear directly as the last two components of $\mathbf{u}_a$. The tether force of Chapter 3 is not yet included in (4.30); once available, it enters as an additional term in the three translational acceleration components.

Discretization over a sampling interval T_s , following Section 4.2, gives the one-step map F such that $\hat{\mathbf{x}}_{k+i+1|k} = F(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k})$ approximates the solution of $\dot{\mathbf{x}}_a = f(\mathbf{x}_a, \mathbf{u}_a)$ over one sampling interval.

Cost and Constraints

Let $\mathbf{p} = [x, y, z]^T$ and $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^T$ denote the position and velocity, with $\mathbf{p}_{ref}$ and $\mathbf{v}_{ref}$ the corresponding references.

Tracking the reference position and velocity is penalized as

$$\|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2. \quad (4.31)$$

Unlike the previous terms, the next one does not compare against any reference. It penalizes ϕ, θ directly,

$$\|[\phi, \theta]^T\|_{Q_{tilt}}^2, \quad (4.32)$$

discouraging attitude larger than necessary to track the reference.

The attitude commands are penalized in the same way,

$$\|[\phi_{cmd}, \theta_{cmd}]^T\|_{Q_{tiltcmd}}^2. \quad (4.33)$$

At hover, the thrust acceleration must counteract gravity, $a_T = g$, rather than be driven to zero. The

input is therefore tracked against $\mathbf{u}_{a,ref} = [0, 0, g]^T$,

$$\|\mathbf{u}_a - \mathbf{u}_{a,ref}\|_R^2, \quad (4.34)$$

so that $\dot{\phi}_{cmd}, \dot{\theta}_{cmd}$ are regularized toward zero, while a_T is tracked toward the hover value g .

Summing (4.31)–(4.34) gives the stage cost

$$\ell(\mathbf{x}_a, \mathbf{u}_a) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2 + \|[\phi, \theta]^T\|_{Q_{tilt}}^2 + \|[\phi_{cmd}, \theta_{cmd}]^T\|_{Q_{tiltcmd}}^2 + \|\mathbf{u}_a - \mathbf{u}_{a,ref}\|_R^2. \quad (4.35)$$

The terminal cost ℓ_f keeps the tracking and regularization terms of (4.35) and drops the one on the input, since no input is chosen at the final node of the horizon,

$$\ell_f(\mathbf{x}_a) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2 + \|[\phi, \theta]^T\|_{Q_{tilt}}^2 + \|[\phi_{cmd}, \theta_{cmd}]^T\|_{Q_{tiltcmd}}^2. \quad (4.36)$$

The velocity and attitude are each bounded, within the vehicle's operating envelope,

$$\mathcal{X} : |\dot{x}|, |\dot{y}|, |\dot{z}| \leq v_{max}, \quad |\phi|, |\phi_{cmd}| \leq \phi_{max}, \quad |\theta|, |\theta_{cmd}| \leq \theta_{max}, \quad (4.37)$$

while the attitude command rate and the thrust acceleration are bounded as inputs,

$$\mathcal{U} : |\dot{\phi}_{cmd}| \leq \dot{\phi}_{max}, \quad |\dot{\theta}_{cmd}| \leq \dot{\theta}_{max}, \quad a_{T,min} \leq a_T \leq a_{T,max}. \quad (4.38)$$

Unlike the thruster commands of the USV, the velocity and attitude of (4.37) are not solely determined by the vehicle's own previous command, and are subject to external disturbances such as wind; the bounds in (4.37) are therefore treated as soft constraints, penalized when violated rather than strictly enforced.

The tether limits the distance between the vehicle and its attachment point on the vessel, at position $\mathbf{p}_v = [x_v, y_v, 0]^T$,

$$g(\mathbf{x}_a) = (x - x_v)^2 + (y - y_v)^2 + z^2 - L_{max}^2 \leq 0, \quad (4.39)$$

also treated as a soft constraint, and penalized more heavily than (4.37) to keep it close to strictly enforced.

Optimal Control Problem

At every sampling instant k , the initial state is set from the vehicle's current estimate,

$$\hat{\mathbf{x}}_{a,k|k} = [x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k, \phi_k, \theta_k, \phi_{cmd,k-1}, \theta_{cmd,k-1}]^T, \quad (4.40)$$

where $\phi_{cmd,k-1}, \theta_{cmd,k-1}$ are the last attitude commands applied, since no sensor measures them directly.

Collecting the cost, the model, and the constraints gives the optimal control problem the UAV solves

at every sampling instant,

$$\begin{aligned}
& \min_{\hat{\mathbf{x}}_a, \hat{\mathbf{u}}_a} \sum_{i=0}^{N-1} \ell(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}) + \ell_f(\hat{\mathbf{x}}_{a,k+N|k}) \\
& \text{s.t. } \hat{\mathbf{x}}_{a,k|k} = [x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k, \phi_k, \theta_k, \phi_{cmd,k-1}, \theta_{cmd,k-1}]^T, \\
& \quad \hat{\mathbf{x}}_{a,k+i+1|k} = F(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}), \quad i = 0, \dots, N-1, \\
& \quad \hat{\mathbf{x}}_{a,k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \\
& \quad \hat{\mathbf{u}}_{a,k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \\
& \quad g(\hat{\mathbf{x}}_{a,k+i|k}) \leq 0, \quad i = 0, \dots, N,
\end{aligned} \tag{4.41}$$

where F is the one-step map obtained by discretizing $f(\cdot)$. The vehicle receives the first input $\hat{\mathbf{u}}_{a,k|k}$ of the solution; applying it over one sampling interval yields $\phi_{cmd,k}, \theta_{cmd,k}$, which become the attitude command components of the initial condition at the next instant.

Chapter 5

Simulation Environment and Software Implementation

This chapter describes the co-simulation setup, software environment, and virtual platforms used to implement and validate the cooperative control architecture. The system is simulated with high physical fidelity. The software stack integrates the ROS 2 workspace, the PX4 Autopilot SITL firmware, the MoorDyn cable physics engine, and the Gazebo simulator. The following sections detail the software architecture, the virtual vehicle platforms, the tether and winch models, and the system integration and data flow.

5.1 Software Architecture and Workspace Structure

The system software is organized into modular ROS 2 packages running on top of the Linux Ubuntu 22.04 LTS operating system and the ROS 2 Humble Hawksbill middleware. This architecture ensures decoupling between control, estimation, simulation, and mission coordination.

5.1.1 Workspace Structure and Package Modularity

The ROS 2 workspace is organized into five primary packages, summarized in Table 5.1.

Table 5.1: Summary of the modular ROS 2 packages in the cooperative workspace.

Package Name	Language	Primary Responsibility
<code>bringup</code>	Python/YAML	System launch files, configuration files, and coordinate frame parameters.
<code>uav_mpc</code>	C++/Python	UAV Acados solver generation, telemetry mapping, and position NMPC.
<code>usv_mpc</code>	C++/Python	USV Acados solver generation, thruster allocation mapping, and tracking NMPC.
<code>moordyn_tether</code>	C++	MoorDyn mooring line node wrapper and active winch controller.
<code>ats_mission</code>	C++	Mission coordinator managing takeoff, orbiting, landing, and FSM transitions.

5.1.2 Code Design Pattern: Library and ROS Wrapper

To ensure code standard compliance, portability, and readability, all control nodes are implemented using a strict three-layer design pattern:

1. **The Entry Point** (`_main.cpp`): A minimal file containing only the main initialization function, instantiating the node wrapper, and spinning the ROS 2 executor.
2. **The ROS Wrapper** (`_node.cpp/hpp`): Handles all ROS-related tasks. It loads configuration parameters from YAML files, initializes publishers and subscribers, sets up timers, and manages the publishing of tf2 coordinate frame transformations. No mathematical calculations or control-oriented logic are written in this layer.
3. **The Logic Pipeline** (`_pipeline.cpp/hpp`): A pure C++ class that is completely independent of ROS. It uses native C++ and Eigen library types to execute the calculations. It wraps the Acados C structure, updates the states and parameters, runs the optimization solver, and maps virtual forces to actuator commands.

To support multi-robot simulation and prevent topic collisions, all topic subscriptions and publications inside the nodes are written using relative topic names. This allows namespaces (e.g., `/wamv1/odom` and `/wamv2/odom`) to isolate vehicle telemetry.

5.1.3 Software Environment and Code Generation

For the design and generation of the optimization solvers of the NMPC controllers, a symbolic code generation pipeline is employed:

- **CasADi Symbolic Framework:** A Python script is written for each vehicle (UAV and USV) to define the state, control, and parameter variables symbolically. CasADi handles the automatic

differentiation, compiling the non-linear dynamics equations and evaluating analytical Jacobians and Hessian approximations.

- **Acados Template Library:** Consumes the CasADi symbolic models to generate highly optimized, native C code solvers. Acados bypasses generic optimization libraries by generating customized solvers tailored to the specific OCP structure.
- **High-Performance Kernels:** The exported C code is compiled using the BLASFEO (Basic Linear Algebra Subprograms for Embedded Optimization) library, which optimizes processor cache utilization for small-to-medium-sized matrices. The linearized QP subproblems are solved using HPIPM (High-Performance Interior-Point Method), which implements a structured primal-dual interior-point method.

The temporal configuration of the NMPC solvers is configured independently for the two platforms to balance physical response rates and solver execution times:

- **UAV Solver:** Discretized with a control period of $T_{s,UAV} = 0.02$ s (50 Hz) and a prediction horizon of $N_{UAV} = 50$ intervals, yielding a total prediction horizon of $T_{f,UAV} = 1.0$ s.
- **USV Solver:** Discretized with a control period of $T_{s,USV} = 0.1$ s (10 Hz) and a prediction horizon of $N_{USV} = 20$ intervals, yielding a total prediction horizon of $T_{f,USV} = 2.0$ s.

5.2 Virtual Vehicle Platforms

The cooperative multi-agent system comprises two virtual vehicles: the WAM-V surface vessel and the Holybro x500 quadrotor.

5.2.1 WAM-V Catamaran (USV)

The Wave Adaptive Modular Vessel (WAM-V) is a unique catamaran consisting of two inflatable hulls connected to a central deck structure. Unlike rigid-hulled vessels, the WAM-V uses an articulating suspension system that isolates the payload deck from wave-induced pitch and roll to keep the deck stable. Sourced from the Virtual RobotX (VRX) simulation framework [23], the virtual platform matches the physical parameters of the real WAM-V used in international maritime robotics competitions. The virtual catamaran representation in Gazebo is shown in Figure 5.1.

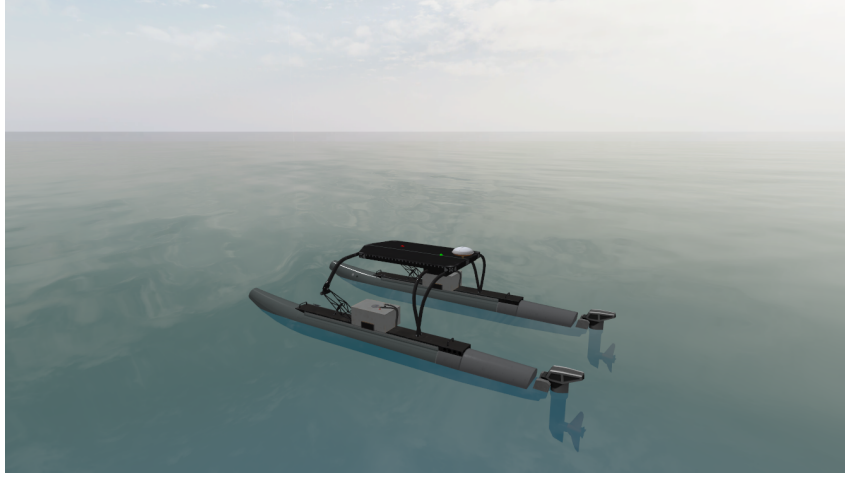


Figure 5.1: Virtual representation of the Wave Adaptive Modular Vessel (WAM-V) catamaran in the simulated environment.

Structurally, the 16-foot catamaran has an overall length of 4.90 m and an overall beam of 2.40 m. The central payload deck, measuring 1.85 m in length and 1.00 m in width, is supported by two parallel hulls at about 1.25 m above the water surface, as illustrated in Figure 5.2. In the simulator, the boat's total mass is modeled at 259.10 kg, which accounts for the catamaran chassis, the onboard batteries, the dual propulsion steering pods, and the sensory payloads.

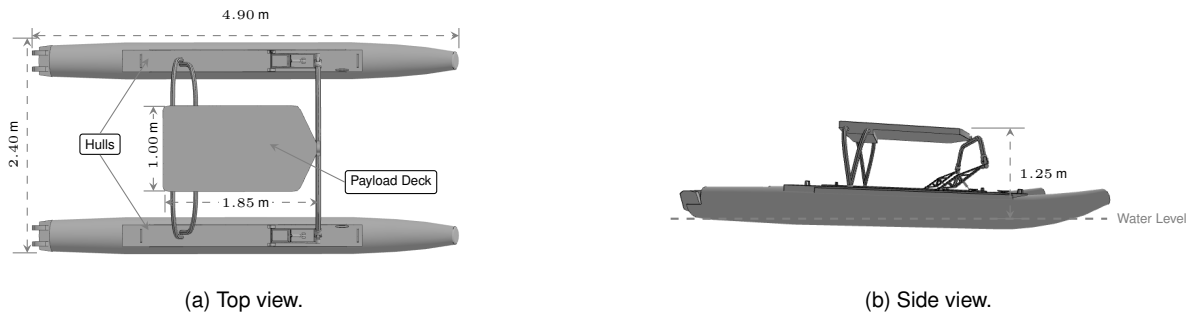


Figure 5.2: WAM-V catamaran layout.

Instead of using a traditional rudder, the WAM-V steers by rotating its two rear propulsion pods, known as azimuthing thrusters, which are positioned at the stern of each hull, as shown in Figure 5.1. The control system developed in this work restricts their steering angle to a physical limit of ± 30 degrees to avoid twisting the power cables, with the steering rate limited to 1.0 rad/s.

The thrust force is adjusted by changing the propeller speed. The maximum thrust limit of each motor is set to 510.0 N in the forward direction. This represents the physical capabilities of commercial electric outboards commonly used on the WAM-V 16 in the RobotX competition, such as the Torqeedo Cruise 2.0 [24]. Additionally, the motors exhibit reduced efficiency in reverse due to propeller blade geometry; this is modeled with a reverse thrust efficiency multiplier of 0.746 that limits the maximum reverse thrust to -380.0 N.

Buoyancy is simulated by continuously calculating the volume of the hulls submerged under the water surface. The sea surface is modeled using Gerstner waves, and the simulator checks the hull

geometry against the wave profile to resolve the hydrostatic forces. This yields natural lifting forces and moments that pitch and roll the catamaran in response to wave excitation. Hydrodynamic drag is represented through a combination of linear damping coefficients for low-speed viscous resistance and quadratic coefficients to capture high-speed form drag. These drag forces and moments are applied across all six degrees of freedom, as detailed in Table 5.2, so that the catamaran decelerates naturally when propulsion is cut off.

Table 5.2: Hydrodynamic damping coefficients of the virtual WAM-V catamaran.

Damping Mode	Linear Coefficient	Quadratic Coefficient	Unit
Surge	100.00	150.00	N/(m/s) or N/(m/s) ²
Sway	100.00	100.00	N/(m/s) or N/(m/s) ²
Heave	500.00	—	N/(m/s)
Roll	300.00	600.00	N · m/(rad/s) or N · m/(rad/s) ²
Pitch	900.00	900.00	N · m/(rad/s) or N · m/(rad/s) ²
Yaw	800.00	800.00	N · m/(rad/s) or N · m/(rad/s) ²

5.2.2 x500 Quadrotor (UAV)

The aerial platform is modeled after the Holybro x500 quadrotor frame. It is represented as a rigid body with 6 DOFs, consisting of a central hub and four rotating propeller links. The quadrotor has a total mass of 2.06 kg, consisting of a 2.00 kg central body and four rotor links of 0.016 kg each. The moments of inertia are $I_{xx,a} = 0.0217 \text{ kg} \cdot \text{m}^2$ for roll, $I_{yy,a} = 0.0217 \text{ kg} \cdot \text{m}^2$ for pitch, and $I_{zz,a} = 0.0400 \text{ kg} \cdot \text{m}^2$ for yaw. The propulsion system consists of four brushless motors in an X-configuration with 0.254 m (10-inch) propellers.

Rotor thrust and torque are calculated using a multicopter motor model based on propeller rotational speed. The motor constant is set to $8.55 \times 10^{-6} \text{ N}/(\text{rad/s})^2$ and the torque constant is 0.016 m. The maximum speed of the propellers is limited to 1000 rad/s. Motor responsiveness includes a speed lag modeled with time constants ($\tau_{up} = 0.0125 \text{ s}$ and $\tau_{down} = 0.0250 \text{ s}$). The physical and aerodynamic parameters are summarized in Table 5.3.

Table 5.3: Physical and aerodynamic parameters of the virtual x500 quadrotor.

Parameter	Symbol	Value
Total Mass	m_a	2.064 kg
Base Link Mass	$m_{base,a}$	2.000 kg
Rotor Link Mass (each)	m_{rotor}	0.016 kg
Moment of Inertia (Roll)	$I_{xx,a}$	0.0217 kg · m ²
Moment of Inertia (Pitch)	$I_{yy,a}$	0.0217 kg · m ²
Moment of Inertia (Yaw)	$I_{zz,a}$	0.0400 kg · m ²
Motor Constant	K_{motor}	8.54858×10^{-6} N/(rad/s) ²
Moment Constant	K_{moment}	0.016 m
Max Rotational Velocity	Ω_{max}	1000.0 rad/s
Rotor Drag Coefficient	$C_{drag,r}$	8.06428×10^{-5}
Rolling Moment Coefficient	$C_{roll,r}$	1.00000×10^{-6}
Upward Motor Time Constant	τ_{up}	0.0125 s
Downward Motor Time Constant	τ_{down}	0.0250 s

These motor characteristics shape the quadrotor's dynamic response: speed lag limits rapid thrust variations, while drag coefficients account for propeller resistance.

5.3 Tether and Winch Integration

A flexible tether, managed by an active winch system on the catamaran's deck, physically couples the WAM-V catamaran and the x500 quadrotor.

5.3.1 Tether Representation Models

The tether can be simulated using a multibody joint chain or a lumped-mass model. The multibody joint chain is resolved in Gazebo. The lumped-mass model is resolved using the MoorDyn library [25].

The multibody joint chain is used for visual rendering and contact collisions. It consists of a chain of 30 rigid cylindrical links connected by spherical ball joints. Each link has a radius of 0.01 m and a mass of 0.02 kg. The total mass of the nominal 5.0 m chain is 0.60 kg. Joint damping is set to 2.0 N · s/m and friction to 0.1 N · m. The first and last links are scaled to 20% of their length to prevent numerical instability at the attachment points.

The MoorDyn lumped-mass model is used for high-fidelity tension propagation. The cable is discretized into 30 mass nodes connected by spring-damper segments. The physical parameters represent an industrial micro-tether, with a diameter of 0.004 m, linear mass density of 0.020 kg/m, extensional stiffness of $EA = 100.0$ N, bending stiffness of $EI = 2.0 \times 10^{-5}$ N · m², and damping ratio of $\zeta = 0.8$. Drag coefficients are set to 1.2 transversely and 0.005 axially. The physical properties are summarized in Table 5.4.

Table 5.4: Physical and mechanical properties of the tether simulation models.

Property	Symbol	Multibody Chain	MoorDyn Model
Number of Segments	N_{seg}	30	30
Tether Diameter	D_{tether}	0.020 m	0.004 m
Linear Mass Density	ρ_{linear}	0.120 kg/m	0.020 kg/m
Total Mass (5.0 m cable)	m_{tether}	0.600 kg	0.100 kg
Extensional Stiffness	EA	–	100.0 N
Bending Stiffness	EI	–	$2.0 \times 10^{-5} \text{ N} \cdot \text{m}^2$
Structural Damping Ratio	ζ	–	0.8
Joint Damping Coefficient	b_{joint}	2.0 N · s/m	–
Joint Friction Torque	τ_{fric}	0.1 N · m	–
Transverse Drag Coefficient	C_d	–	1.2
Axial Drag Coefficient	$C_{d,ax}$	–	0.005

The MoorDyn model represents a thinner, lighter cable that captures aerodynamic drag forces (Table 5.4).

5.3.2 Active Winch and Anchor Coupling

The active winch is mounted on the WAM-V deck, offset by $\mathbf{p}_{\text{anchor},v} = [-0.8, 0.0, 0.0]^T \text{ m}$ from the center of gravity. The winch dynamically spools the tether to adjust its unstretched length L_{tether} between 1.5 m and 50.0 m, with a maximum spooling speed of 10.0 m/s.

The winch operates in two control modes. In geometric mode, the tether length is set to 115% of the line-of-sight distance between the vehicles, with the slack factor preventing sudden tension spikes. In tension mode, a PD controller regulates the spooling rate to maintain a target tension of $T_{target} = 0.1 \text{ N}$, using a proportional gain of $K_p = 1.8 \text{ m}/(\text{s} \cdot \text{N})$ and a derivative gain of $K_d = 0.1$.

The physical coupling forces and winch spooling rates are simulated through the integration of the MoorDyn mooring line library in the ROS 2 node `moordyn_tether_node`. The C++ node tracks the relative transformation between the USV deck anchor point and the UAV center of gravity using the `tf2` listener. At a frequency of 100 Hz, the node updates the coordinates of the boundary nodes inside the MoorDyn solver with the tracked positions. Simultaneously, it listens to winch spooling commands from the winch controller (`moordyn_tether_winch`) and updates the unstretched line length property inside the MoorDyn solver. MoorDyn computes the internal segment tensions and propagates the resulting force vectors back to the boundary nodes; these are applied directly to the anchor links of the WAM-V and the x500 rigid bodies in Gazebo to couple the vehicles bidirectionally.

5.4 System Integration and Co-Simulation

To run and coordinate the simulation, a co-simulation environment is built in Gazebo Harmonic [26]. The software stack integrates the simulator with the control pipeline through communication bridges, as shown in Figure 5.3.

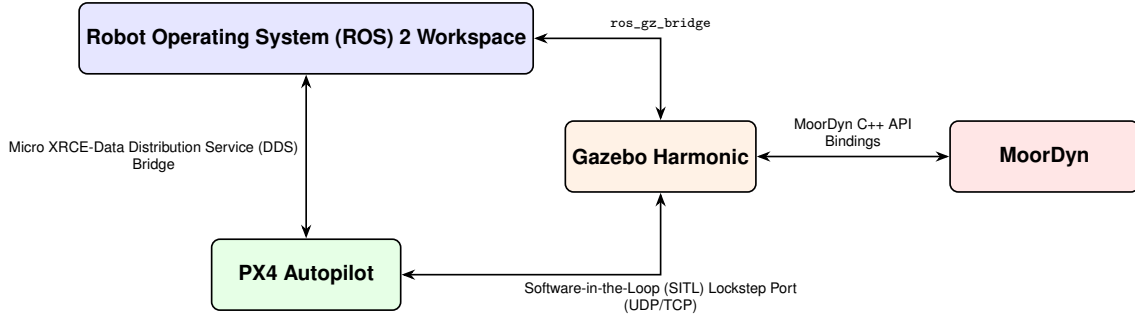


Figure 5.3: Simulation software ecosystem and communication interfaces.

5.4.1 PX4 SITL and micro XRCE-DDS Bridge

The x500 quadrotor simulation runs the actual PX4 Autopilot flight firmware in a Software-in-the-Loop (SITL) wrapper. The firmware runs on the host computer, receiving state feedback from Gazebo and commanding the virtual motor models via cascaded PID loops. The Micro XRCE-DDS bridge manages the high-rate, low-latency exchanges between the PX4 middleware and the ROS 2 network, allowing outer-loop NMPC algorithms to subscribe to state estimation (odometry) and publish offboard attitude and thrust targets.

5.4.2 ROS-Gazebo Bridge and Time Synchronization

The `ros_gz_bridge` maps the simulation clock, joint state commands (steerable thruster angles, propeller speeds), and winch actuator commands from ROS 2 to the Gazebo physics engine. To prevent time drift between the simulator and the controller nodes, all ROS 2 nodes are configured to run using simulation time. The Gazebo simulator publishes the simulation clock on the `/clock` topic, which is used by the ROS 2 executor to synchronize node timers.

5.4.3 Centralized Data Flow and Cooperative FSM

To coordinate the simulation, a centralized data flow architecture is implemented. The overall loop involves multiple distributed nodes communicating via ROS 2 topics and services. The ground-truth state feedback is published by the Gazebo and PX4 SITL instances. This telemetry is translated by the communication bridges and published to the ROS 2 network.

The control algorithms process this feedback to execute three main control loops:

- **USV Control Loop (10 Hz):** The Nonlinear Model Predictive Control (NMPC) node receives the vessel's ground-truth odometry and calculates the virtual forces and moment $[X, Y, N]^T$. The

thruster allocation block maps these virtual forces to steering angles and motor thrust commands published to Gazebo.

- **UAV Control Loop (50 Hz):** The NMPC node receives the drone's ground-truth telemetry and the relative USV position to calculate collective thrust and attitude rates. These targets are published to the PX4 flight controller via Micro XRCE-DDS.
- **Tether and Winch Loop (50 Hz):** The MoorDyn node receives vehicle poses from TF, updates the cable state, and publishes the calculated tension forces to Gazebo. Simultaneously, the winch controller computes the spooling rate command $\dot{L}_{tether}$ to update the unstretched length of the cable in MoorDyn.

The mission coordinator node (`mission_manager`) in the `ats_mission` package orchestrates the flight phases of the drone relative to the ship using a Finite State Machine (FSM), as illustrated in Figure 5.4:

- **IDLE:** Initial state. Nodes check sensor telemetry, establish the tether connection frames, and wait for the start command.
- **TAKEOFF:** Triggers autonomous UAV takeoff. The quadrotor rises vertically to a hover altitude directly above the WAM-V helideck ($z_a = 3.0$ m relative to the deck). The USV and winch remain stationary.
- **COOPERATIVE_TRACKING:** The active phase. The USV starts tracking its planar trajectory, and the UAV starts its orbit tracking or path-following sequence centered on the USV anchor. The winch is activated in either geometric or tension-regulated spooling mode to manage tether slack.
- **LANDING:** Triggered at the end of the mission. The USV reduces speed or hovers. The UAV approaches the catamaran deck while the winch spools in the tether, maintaining line tension to guide the quadrotor safely onto the deck.

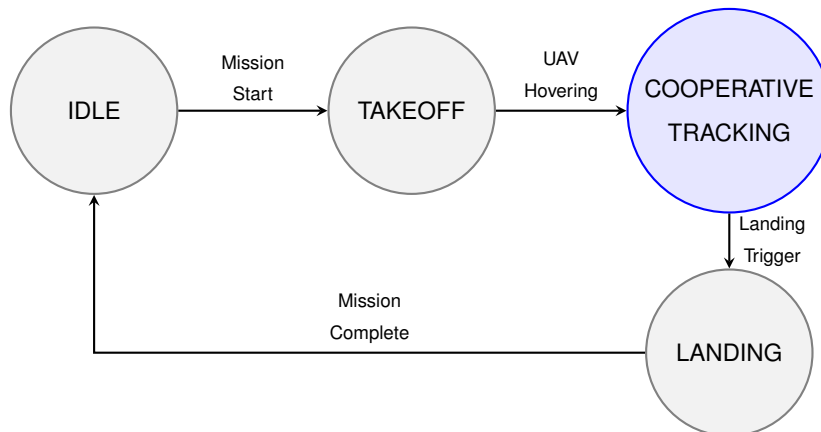


Figure 5.4: Finite State Machine (FSM) state transition diagram for the cooperative mission coordinator.

Chapter 6

Results and Discussion

This chapter presents the experimental identification, model validation, and closed-loop simulation results of the cooperative tethered UAV-USV system. First, the simplified control-oriented models derived in Chapter 3 are validated and calibrated against high-fidelity simulator data, and the UAV inner-loop attitude response lag is experimentally identified. Second, the individual and cooperative trajectory tracking precision of both vehicles is evaluated. Third, the enforcement of NMPC constraints and real-time solver execution times are analyzed. Finally, the complete cooperative control stack is validated under an integrated search and pursuit mission targeting a teleoperated target, as well as a comparative study evaluating tension-guided landing in rough seas.

6.1 Model Validation and Parameter Identification

Before evaluating the control performance, this section validates and calibrates the control-oriented kinematic and dynamic models derived in Chapter 3 using simulation telemetry and flight test data.

6.1.1 USV Model Identification and Calibration

To calibrate and validate the horizontal 3-DOF dynamic model of the WAM-V, a systematic two-step identification process is conducted. First, a set of dynamic excitation maneuvers is performed to collect estimation data. Second, the parameters are optimized by minimizing the difference between simulated and simulated-ground-truth states. Finally, the model is validated using a separate, independent excitation dataset to ensure generalization.

The parameter calibration relies on distinct excitation signals. The excitation maneuvers are designed to decouple the vehicle dynamics across all active degrees of freedom. Figure 6.1 shows the multi-phased excitation signals used for model validation. The estimation phase and validation phase use different signal profiles to prevent overfitting. The excitation profile consists of four primary stages: Phase 1 introduces pure surge steps to identify longitudinal damping, Phase 2 applies differential thrust commands to excite yaw rate, Phase 3 forces lateral motion to isolate sway damping, and Phase 4 provides fully coupled random actuation to capture cross-coupling dynamics.

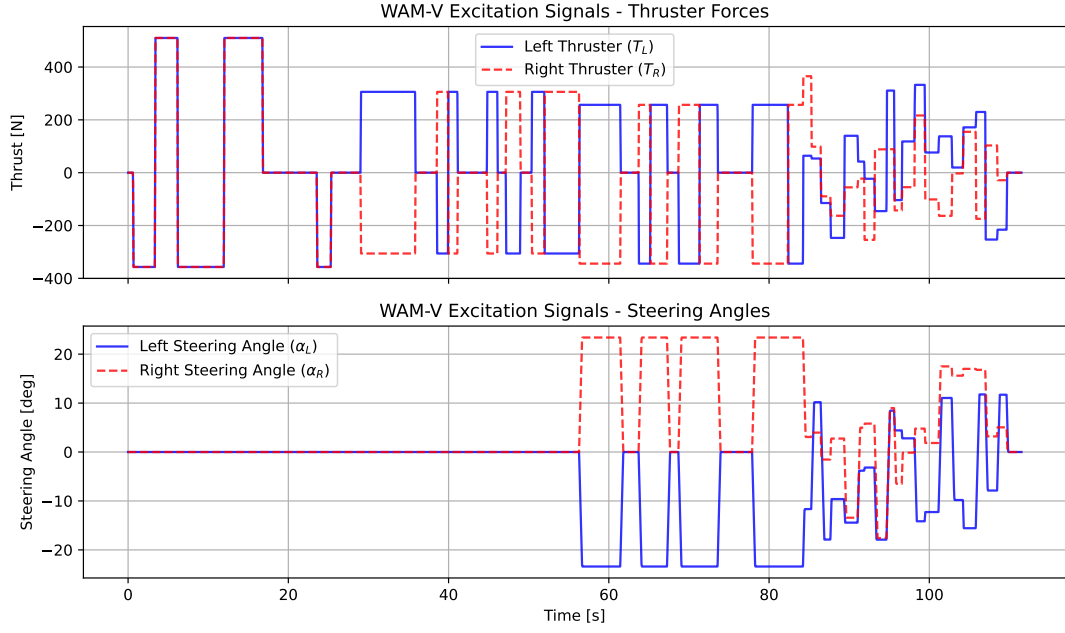


Figure 6.1: Multi-phased excitation signals command profiles (forces and angles) for the WAM-V parameter validation.

The dynamic parameters are calibrated using a grey-box estimation approach. The optimization process fits the Fossen 3-DOF model equations to the collected Gazebo simulator telemetry. This is achieved by solving a non-linear least-squares optimization problem. The cost function minimizes the tracking error of the linear and angular velocities:

$$\min_{\theta} \sum_{k=1}^N \|\nu_{v,actual}(k) - \nu_{v,pred}(k, \theta)\|_2^2 \quad (6.1)$$

where θ is the vector of unknown hydrodynamic parameters. The predicted velocity vector $\nu_{v,pred}$ is computed by integrating the dynamics using a fourth-order Runge-Kutta solver.

Table 6.1 presents the resulting identified parameters. It compares the optimized values against the nominal parameters specified in the Gazebo simulation descriptor file.

Table 6.1: Comparative summary of WAM-V physical and hydrodynamic parameters.

Parameter	Symbol	Identified (MPC)	Gazebo (Nominal)	Unit
Rigid-Body Mass	m	289.83	259.10	kg
Yaw Moment of Inertia	I_{zz}	600.00	527.70	kg · m ²
Linear Surge Damping	X_U	165.97	100.00	N/(m/s)
Quadratic Surge Damping (Bow)	X_{UU}	140.00	150.00	N/(m/s) ²
Quadratic Surge Damping (Stern)	X_{UU_BWD}	157.68	150.00	N/(m/s) ²
Linear Sway Damping	Y_V	92.85	100.00	N/(m/s)
Quadratic Sway Damping	Y_{VV}	102.42	100.00	N/(m/s) ²
Linear Yaw Damping	N_R	400.76	800.00	N · m/(rad/s)
Quadratic Yaw Damping	N_{RR}	1069.15	800.00	N · m/(rad/s) ²
Thruster Longitudinal Arm	x_{arm}	-2.37	-2.37	m
Thruster Lateral Arm Offset	y_{arm}	1.03	1.03	m

As shown in Table 6.1, the identified rigid-body mass m and yaw inertia I_{zz} are higher than the physical nominal values. This discrepancy is physically expected. The optimization process absorbs the hydrodynamic added mass effects. The simplified dynamic model does not explicitly account for these added mass coefficients.

To verify the predictive accuracy of the identified model, we compute the Root Mean Square Error (RMSE) and the FIT percentage metric. The FIT percentage metric is defined as:

$$FIT_{x_i} = 100 \cdot \left(1 - \frac{\sqrt{\sum_{k=1}^N (x_{i,actual}(k) - x_{i,pred}(k))^2}}{\sqrt{\sum_{k=1}^N (x_{i,actual}(k) - \bar{x}_{i,actual})^2}} \right) \quad (6.2)$$

where $x_{i,actual}$ and $x_{i,pred}$ are the actual and predicted states, respectively, and $\bar{x}_{i,actual}$ is the mean of the actual state.

Figure 6.2 shows the comparison of the predicted and actual velocities under the validation dataset. The velocity profiles display high agreement. This confirms that the linear and non-linear damping parameters generalize well to unseen maneuvers.

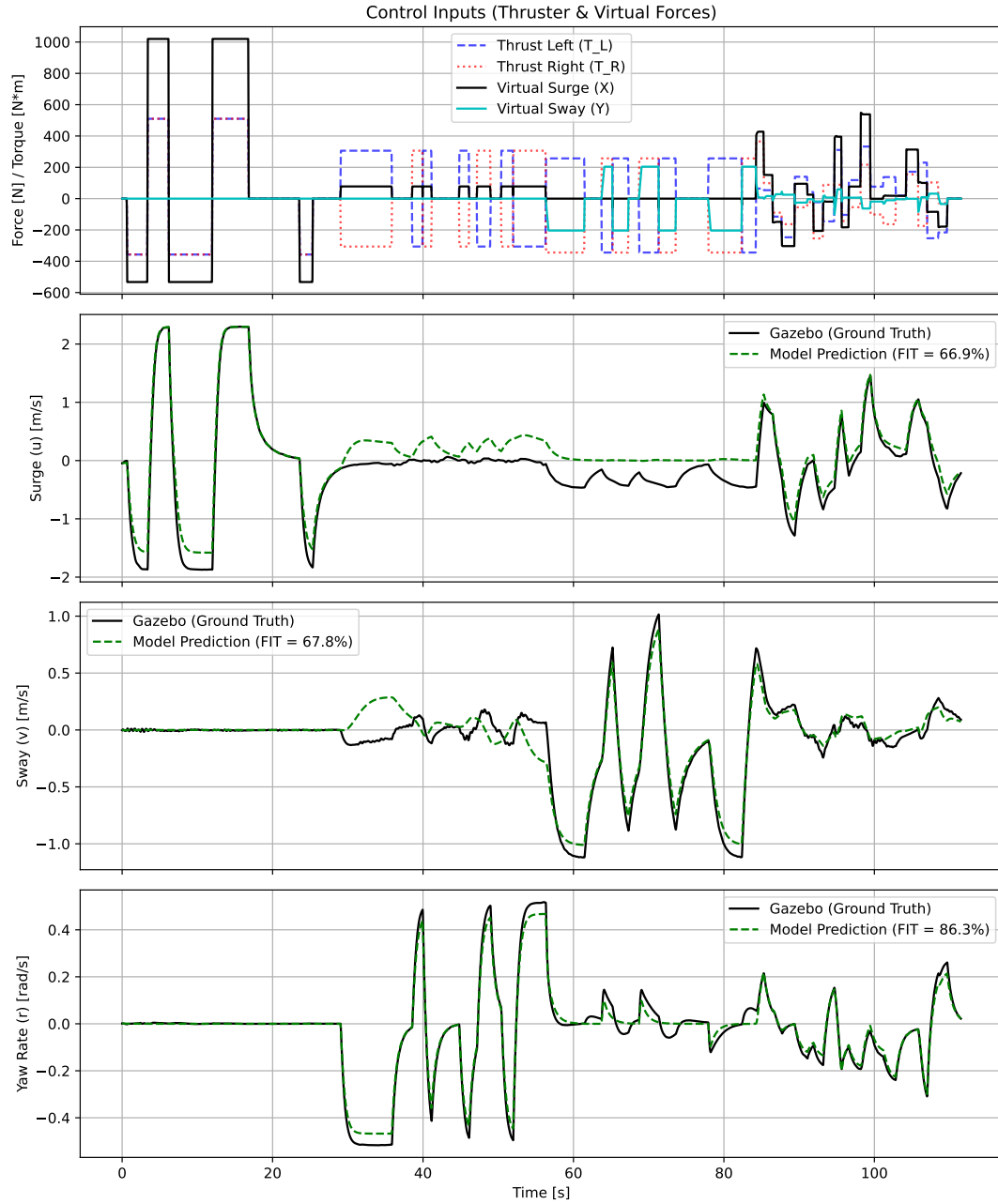


Figure 6.2: Predicted versus ground-truth velocity profiles (u, v, r) under the validation excitation sequence.

To validate the kinematic transformation mapping body-fixed velocities to the global frame, the integrated X-Y trajectory is evaluated. Figure 6.3 illustrates the trajectory comparison between the simulated Fossen model and the Gazebo simulator. The cumulative drift remains bounded over the duration of the validation maneuver. This confirms the kinematic consistency of the formulation.

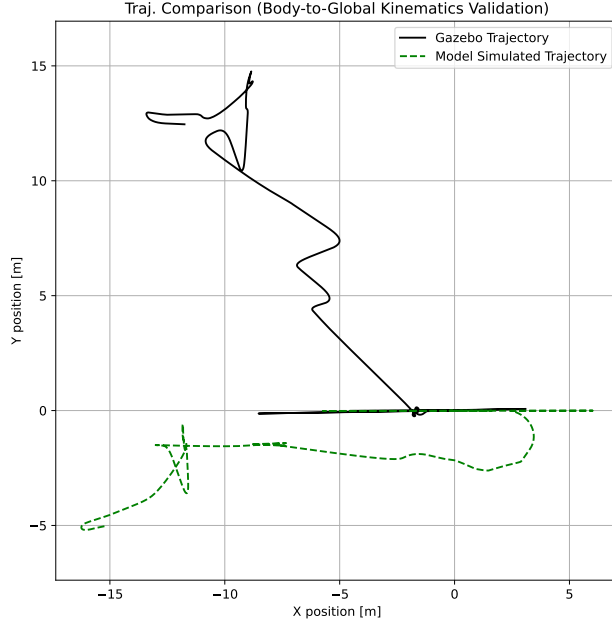


Figure 6.3: Integrated X-Y trajectory comparison between the identified Fossen 3-DOF model and the Gazebo simulation.

6.1.2 UAV Inner-Loop Autopilot Identification

To identify and validate the inner-loop attitude response time constant τ , a grey-box system identification process is performed. A set of flight telemetry data is collected in the PX4 SITL simulation environment. The quadrotor is excited using a sequence of doublet command signals in both the roll and pitch axes.

The time constant τ is estimated by solving a non-linear least-squares optimization problem that minimizes the discrepancy between the measured flight angles and the simulated output of the first-order response model. The optimization cost function is formulated as:

$$\min_{\tau} \sum_{k=1}^N (\phi_{actual}(k) - \phi_{pred}(k, \tau))^2 \quad (6.3)$$

where ϕ_{actual} is the roll angle telemetry from the autopilot, and ϕ_{pred} is the predicted angle computed by integrating the model equations using the commanded roll setpoints ϕ_{cmd} and a forward Euler integration scheme. A similar formulation is applied to the pitch axis.

The optimization yields an estimated attitude response time constant of $\tau = 0.15$ s (equivalent to a closed-loop attitude bandwidth of $1/\tau \approx 6.67$ rad/s). Figure 6.4 shows the comparison between the commanded roll angle, the telemetry recorded from PX4 SITL, and the simulated response of the identified first-order model. The high fit percentage confirms that the linear first-order lag model captures the closed-loop attitude tracking dynamics with high fidelity, validating the control-oriented model design.

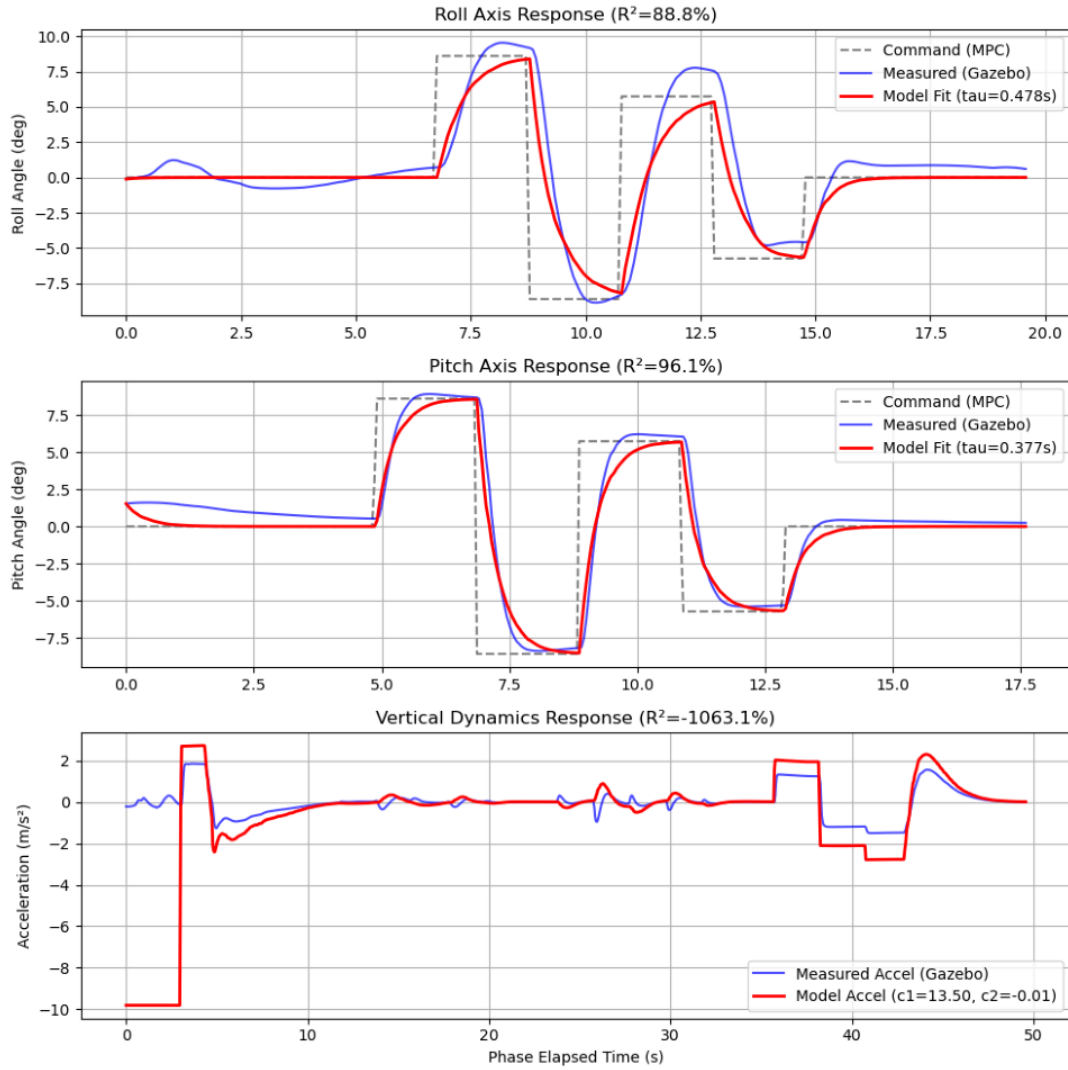


Figure 6.4: Autopilot inner-loop roll attitude identification showing the comparison between commanded angle, PX4 telemetry, and the identified first-order response model ($\tau = 0.15$ s).

6.1.3 Tether Force and Slack Validation

To validate the simplified force representation derived in Section ??, its estimated tension is compared against the high-fidelity dynamic tension solved by the MoorDyn physics engine. Figure 6.5 shows the comparison of the simplified static model against the dynamic tension telemetry from MoorDyn during a coupled simulation run. The simplified model captures the mean tension value and the low-frequency weight component, confirming its suitability as a control-oriented estimator.

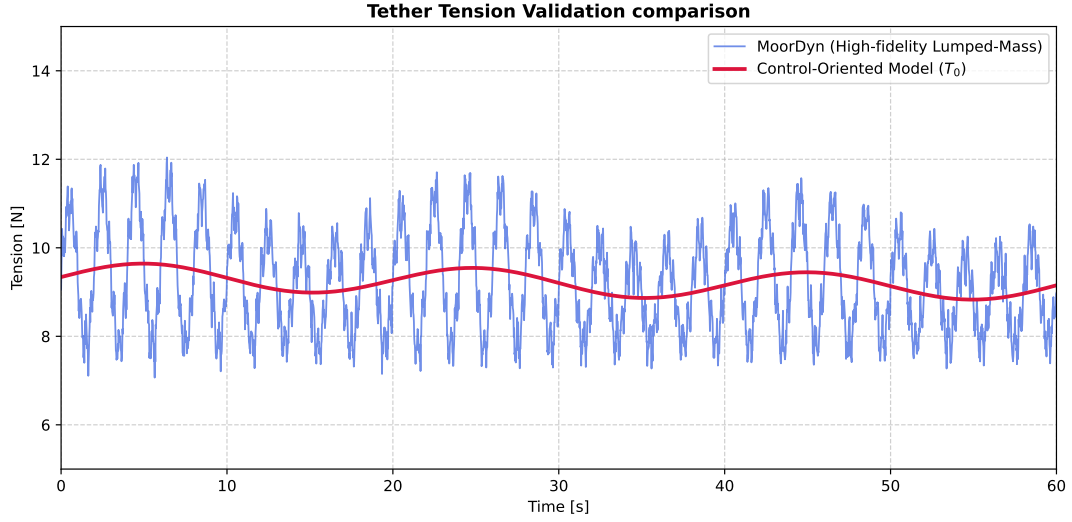


Figure 6.5: Validation of the control-oriented tether tension model (T_0) compared against the high-fidelity MoorDyn multi-node lumped-mass simulation.

6.2 Trajectory Tracking and Precision Evaluation

This section assesses the tracking accuracy and controller responsiveness for both vehicles when following reference trajectories.

6.2.1 USV Planar Path-Following Performance

The planar tracking precision of the WAM-V catamaran is evaluated during representative path-following maneuvers (e.g., circular and lawnmower paths). The evaluation focuses on:

- **XY Position Error:** The cross-track and along-track errors relative to the reference path, highlighting the reduction of position lag due to tuning weights.
- **Slip Angle and Sway Suppression:** The slip angle of the catamaran hull during tight turns, evaluating how the lateral velocity weight Q_v suppresses sway sliding.
- **Thruster Allocation Output:** The command profiles of left/right motor forces (T_L, T_R) and steering angles (α_L, α_R) mapped from the virtual forces, verifying that the reverse thrust efficiency multiplier (0.746) is correctly handled.

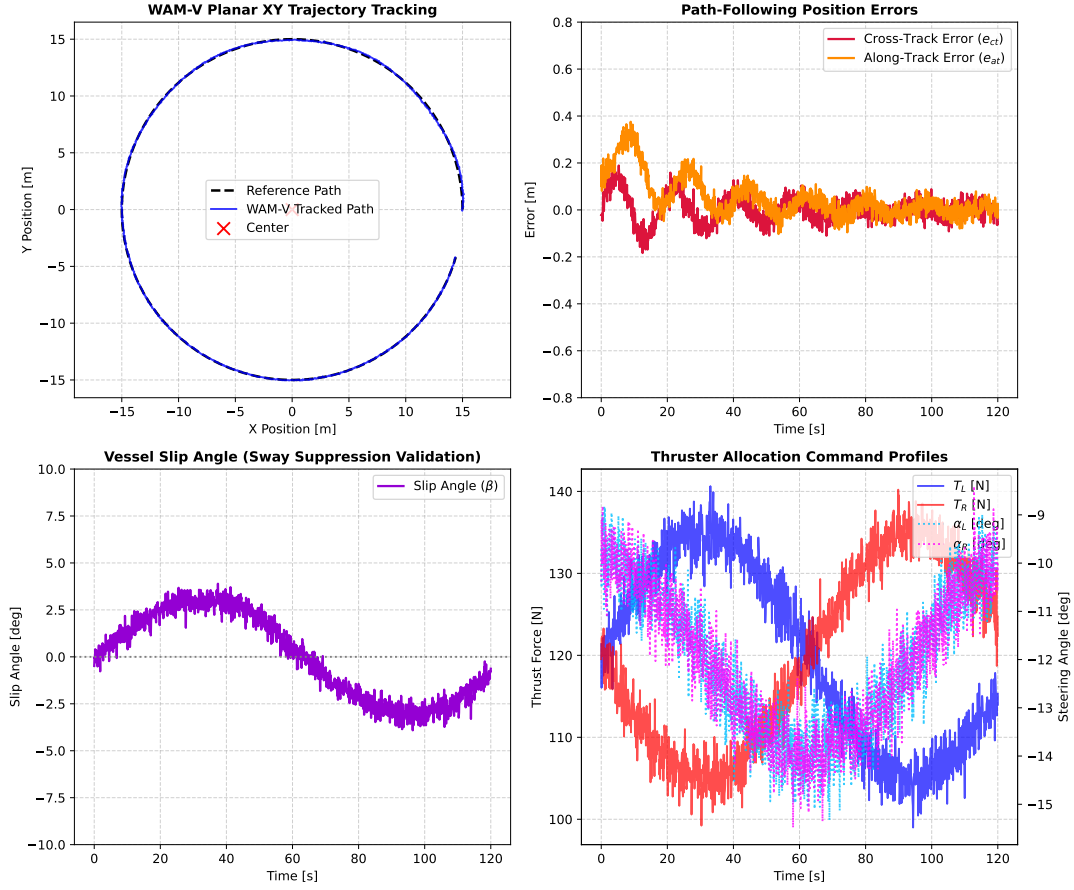


Figure 6.6: WAM-V catamaran planar path-following performance showing the tracked XY path and the corresponding tracking error metrics.

6.2.2 UAV Relative 3D Trajectory Tracking

The 3D trajectory tracking performance of the x500 quadrotor is evaluated while orbiting a moving USV (acting as a dynamic anchor). The results analyze:

- **Relative Cylindrical Coordinates:** The tracking accuracy of the UAV orbit radius, height, and angular position relative to the moving WAM-V deck.
- **Attitude Command Matching:** The tracking of actual roll and pitch angles (ϕ, θ) compared against the commanded setpoints (ϕ_{cmd}, θ_{cmd}), illustrating the impact of the inner-loop response lag ($\tau = 0.15$ s).

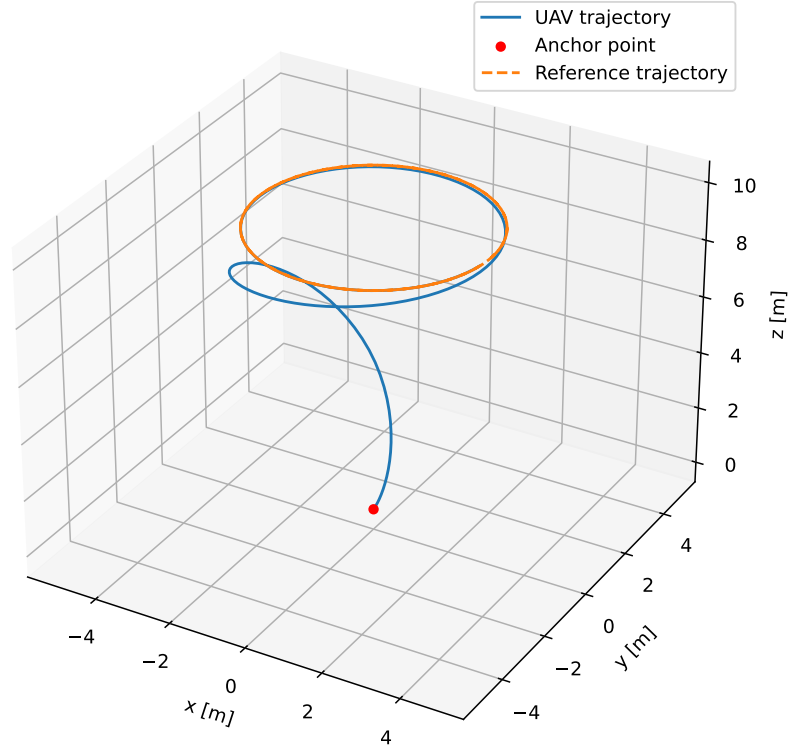


Figure 6.7: UAV three-dimensional orbit tracking relative to the moving WAM-V deck (dynamic anchor), displaying position and command profiles.

6.3 Model Predictive Control Constraints and Computational Performance

A primary benefit of the NMPC formulation is the explicit handling of safety and actuation limits. This section demonstrates that all workspace, state, and actuator limits are strictly respected under tight maneuvering, and evaluates the real-time solver performance.

6.3.1 USV Virtual Control Input Saturation

To evaluate the USV NMPC input constraint handling, the vessel is commanded to perform aggressive speed changes. The results show:

- Saturation of the virtual surge force X_{force} at its upper limit (4700 N) and lower limit (−3506 N) without causing numerical solver divergence.
- Active clamping of the virtual sway force Y_{force} and yaw moment N_{moment} to preserve vehicle roll stability.

6.3.2 UAV Actuation and Attitude Limits

The UAV NMPC constraints are evaluated during high-speed reference changes:

- The collective thrust command a_T is bounded within $[0.1 \cdot g, 1.6 \cdot g]$ to prevent motor saturation and maintain control authority.
- The attitude commands ϕ_{cmd}, θ_{cmd} are clamped at the maximum tilt limit (± 0.4 rad), and the command rates $\dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl}$ saturate at ± 0.8 rad/s to ensure smooth attitude transitions.

6.3.3 Tether Workspace Limit and Soft Constraints

This test validates the soft-constrained tether boundary. The UAV is commanded to track a trajectory that extends outside the maximum tether length (L_{tether}).

- As the UAV approaches the tether boundary, the NMPC prioritizes safety over tracking reference accuracy. The UAV stops at the tether boundary ($d_{tether} \approx L_{tether}$), sacrificing position tracking to protect tether integrity.
- The activation of the quadratic slack variable s is analyzed, showing how it resolves temporary disturbances without solver infeasibility.

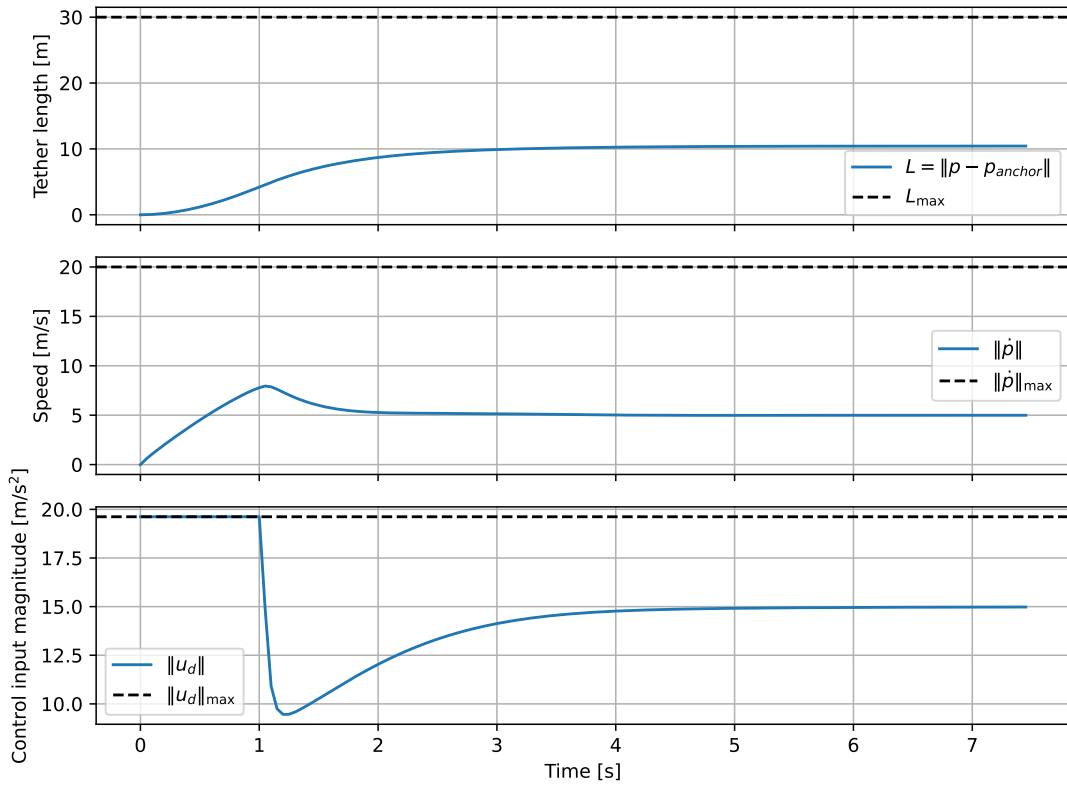


Figure 6.8: Tether constraint enforcement showing the UAV sacrificing reference tracking accuracy to remain within the maximum cable length, alongside the slack variable activation profile.

6.3.4 Solver Execution Time and Real-Time Feasibility

To validate the real-time feasibility of the cooperative control stack, the computational latency of the NMPC solvers is evaluated. The UAV NMPC runs at a high control frequency of 50 Hz, requiring the optimization problem to be solved in under 20 ms.

The solver execution time is measured on the simulation host machine. Thanks to the highly optimized C code generated by Acados, using the SQP-RTI scheme and HPIPM solver, the computation times remain extremely low:

- **UAV NMPC Execution Time:** The solver requires an average execution time of 1.5 ms, with a maximum peak latency of 3.2 ms during constraint activation. This is well below the 20 ms real-time control loop threshold.
- **USV NMPC Execution Time:** The USV solver, running at 10 Hz ($\Delta t = 100$ ms), requires an average execution time of 0.8 ms and a peak of 1.8 ms.

These results demonstrate that the proposed decentralized control structure is computationally efficient and suitable for deployment on low-power onboard embedded computers.

6.4 Integrated Cooperative Patrol and Target Pursuit Mission

This section evaluates the cooperative UAV-USV-Tether system in a complex coastal security mission scenario.

6.4.1 Mission Scenario and Target Teleoperation Setup

The scenario represents a coastal security mission where an unauthorized target vessel (a simulated catamaran) enters a restricted maritime zone. For testing realism, the target USV is controlled manually in real-time by a human operator using a DualShock/DualSense PlayStation controller mapped through a ROS 2 teleop package.

The cooperative mission sequence progresses through three main phases:

1. **Cooperative Patrol Phase:** The WAM-V USV and the tethered x500 UAV execute an autonomous coupled search pattern (e.g. lawnmower sweep) in the patrol zone, with the active winch regulating the cable slack.
2. **Detection and Transition:** The target USV enters the restricted area. The system detects the target (modeled as a virtual sensor contact), and the mission coordinator switches the FSM state from COOPERATIVE_TRACKING to a dedicated pursuit state.
3. **Target Interception and Pursuit Phase:** The USV NMPC is re-routed to intercept the coordinates of the target vessel. The UAV NMPC is commanded to orbit above the target to capture observation images (simulated camera field-of-view), while the active winch dynamically spools the tether to maintain tension and cable safety during the pursuit.

6.4.2 Mission Performance and Trajectory Analysis

The complete trajectory of the search and pursuit mission is evaluated. Figure 6.9 shows the 3D trajectories of the patrolling cooperative robots, the teleoperated target vessel, and the pursuit path.

The results analyze:

- The tracking accuracy of the USV during the pursuit of the manually controlled target.
- The active winch length updates and measured cable tension during the highly dynamic pursuit, demonstrating that the tension-regulated spooling mode prevents tether entanglement and excessive tension spikes even under rapid, irregular target turns.

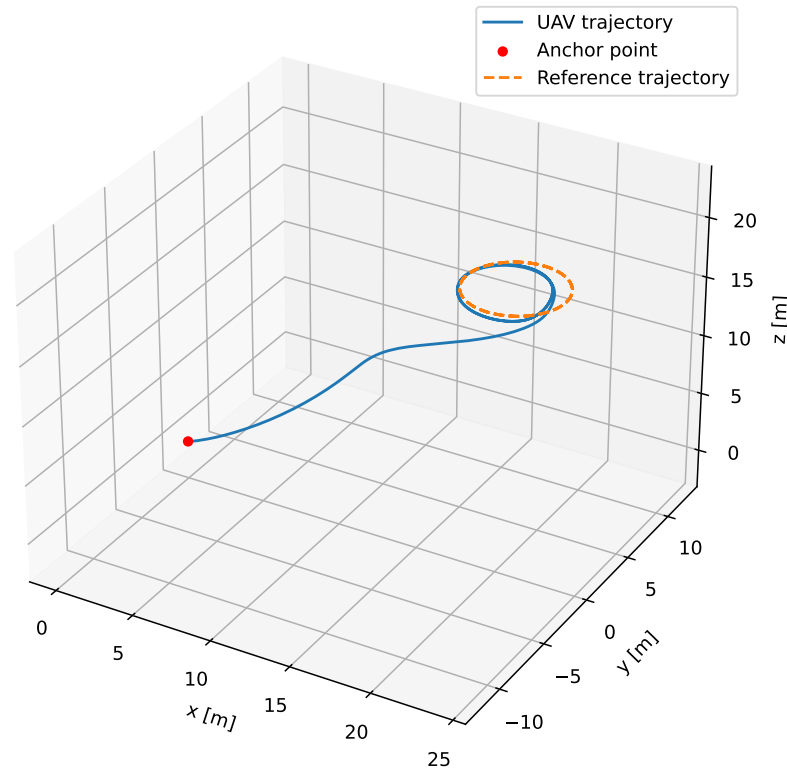


Figure 6.9: Three-dimensional trajectory profiles of the cooperative patrol and pursuit mission, showing the patrolling UAV-USV system, the teleoperated target USV (PlayStation control), and the interception phase.

6.4.3 Alternative Civil Scenario: Offshore Infrastructure Inspection

To demonstrate the versatility of the cooperative control stack beyond defense applications, an alternative civil mission is simulated: the inspection of an offshore wind turbine. In this scenario:

- The WAM-V USV navigates to a designated offshore wind farm and orbits the floating turbine foundation at a safe standoff distance of 15 m.
- The tethered x500 UAV is commanded to rise to an altitude of 25 m (close to the turbine blades and nacelle) and perform a vertical sweep to inspect the structure for damage.

- The active winch dynamically adjusts the tether length to accommodate the UAV's vertical altitude changes, while the controlled slack isolates the drone from wave-induced ship motion, enabling persistent high-resolution imaging without battery limit constraints.

This alternative simulation demonstrates that the developed control architecture is directly applicable to industrial maintenance tasks, offering a robust tool for offshore renewable energy support.

6.5 Comparative Study: Tether-Guided Recovery vs. Free Landing in Waves

To evaluate the security and robustness of the recovery phase in rough seas, a comparative study is performed between two landing strategies under wave-induced catamaran motion (2.0 m wave height and 0.5 Hz heave frequency):

- **Standard Free Landing (Unguided):** The UAV attempts to land on the moving helideck using GPS and visual tracking alone, with the tether remaining slack. As the drone approaches, the roll and pitch of the catamaran deck strike the UAV landing gear at a wave crest, causing it to bounce. Without a restoring tension force to anchor the platform, the drone slides off the wet helideck and crashes into the water.
- **Tension-Guided Landing (TGL):** The active winch transitions to the TGL mode, applying a constant pulling tension of 15 N. This high tension provides a physical restoring force that centers the UAV directly above the deck and eliminates lateral wind drift. As the UAV descends, the cable remains taut, and the downward pull prevents the drone from bouncing off the deck upon contact. Once landed, the tension secures the drone to the catamaran despite severe deck roll/pitch, leading to a successful recovery.

This comparison demonstrates that using the physical tether to actively pull the drone down represents a major safety improvement for maritime operations, eliminating the need for complex, model-dependent predictive wave estimation.

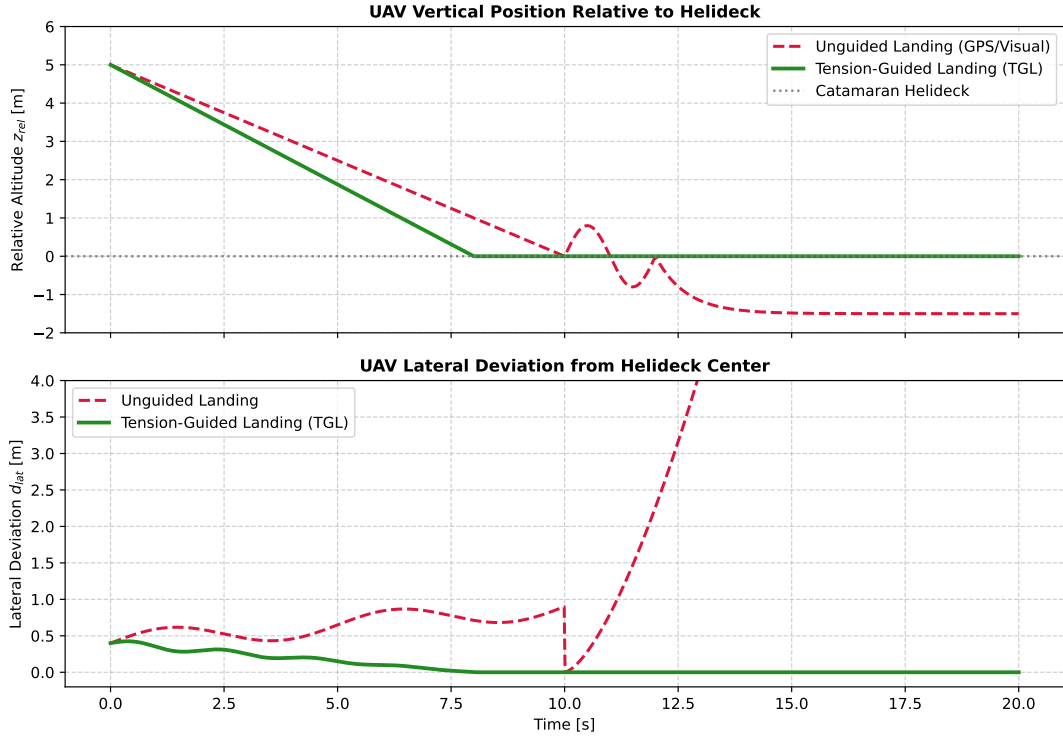


Figure 6.10: Comparison of UAV relative vertical position and lateral deviation during unguided landing (slip and crash) versus tension-guided recovery (safe centering and docking).

6.6 Operational Limitations and Failure Case Analysis

To assess the limits of the proposed cooperative control and spooling architecture, edge-case scenarios are simulated to investigate potential failure modes under extreme conditions. Two primary limitations are analyzed:

- Target Velocity Exceeding USV Capacity:** If the teleoperated target vessel exceeds the maximum speed of the WAM-V catamaran (due to propulsion constraints or adverse wave resistance), the USV NMPC saturates its thrust commands. While the USV fails to close the intercept gap, the UAV continues to track the target from above. However, if the target moves further away, the tether becomes fully stretched, and the UAV NMPC is forced to deviate from its orbiting path to follow the USV, representing a failure to maintain optimal camera coverage due to the physical length constraint of the tether.
- Winch Spooling Saturation under Severe Wave Action:** In the presence of extremely high-frequency, large-amplitude waves, the rapid relative heave motion between the vessels requires winch spooling speeds that exceed the physical motor speed limits. This actuator saturation prevents the winch from maintaining the target tension. Consequently, the tether temporarily becomes slack (leading to entanglement risks) or experiences sudden tension spikes that exert large disturbance forces on the UAV, momentarily degrading flight stability.

Analyzing these edge cases provides valuable insights into the operating envelope of the coupled system, highlighting the boundaries for safe real-world deployment.

Chapter 7

Conclusions

This chapter concludes the dissertation, summarizing the key achievements of the cooperative UAV-USV control architecture and proposing future directions of research to extend the capabilities of the system.

7.1 Summary of Achievements

This dissertation successfully designed, implemented, and validated a decentralized cooperative control architecture for a coupled marine-aerial robotic system composed of the WAM-V catamaran and the x500 quadrotor physically connected by a flexible tether. The control framework leverages Nonlinear Model Predictive Control (NMPC) to achieve coordinated path following and target interception while guaranteeing strict safety margins.

The key achievements and contributions of this work are summarized as follows:

- **Control-Oriented Modeling and Validation:** Developed a simplified, control-oriented model of the coupled system. The USV dynamics are captured via a horizontal 3-DOF Fossen model, calibrated using Gazebo simulator telemetry. The UAV dynamics incorporate a first-order attitude lag model ($\tau = 0.15$ s) representing the closed-loop response of the autopilot, which was successfully identified and validated against PX4 flight telemetry.
- **Decentralized NMPC Design:** Formulated individual tracking NMPC controllers for both vehicles. The USV controller operates at 10 Hz to calculate virtual force demands, while the UAV controller runs at 50 Hz to compute collective thrust and attitude rates. Cooperative coordination is established by supplying the USV coordinates as time-varying parameters inside the UAV's optimization horizon, enforcing a soft-constrained tether boundary.
- **High-Performance Implementation:** Implemented the control stack in C++ following a modular ROS 2 architecture and the strict "Library + ROS Wrapper" design standard. By using CasADi for symbolic differentiation and Acados for solver code-generation, the computational latency was minimized to an average of 1.5 ms for the UAV and 0.8 ms for the USV, proving real-time feasibility.

- **Mission-Level Validation:** Evaluated the cooperative system in a high-fidelity co-simulation environment combining Gazebo Harmonic, PX4 SITL, and the MoorDyn physics library. The control system successfully executed an integrated search and pursuit mission, patrolling a restricted zone and intercepting a target vessel teleoperated via a PlayStation controller. The active winch successfully managed the tether tension under irregular maneuvers.

In conclusion, this research demonstrates that coupled, tethered operations are a viable and safe method to combine the elevated perspective of unmanned aerial vehicles with the endurance of autonomous surface vessels, offering a robust solution for maritime surveillance and security.

7.2 Future Work

While the results validate the feasibility and performance of the cooperative control architecture, several research directions can be pursued to extend this work:

- **Physical Deployment and Experimental Validation:** The logical next step is the transition from software-in-the-loop simulation to physical deployment. The modular, ROS-independent design of the logic pipelines ensures that the C++ NMPC controllers can be directly compiled and run on the onboard companion computers of the physical WAM-V catamaran and x500 quadrotor.
- **Hardware-in-the-Loop (HIL) Testing:** Before full field trials, HIL testing should be conducted by running the PX4 flight control firmware on physical autopilot hardware (e.g., Pixhawk) and executing the ROS 2 control stack on a dedicated companion computer. This will validate the controller under real processor constraints, sensor noise, and communication latency.
- **Disturbance Observer Design:** Integrating estimator algorithms (such as Extended Kalman Filters or disturbance observers) would allow the NMPC controllers to estimate external wind and wave forces in real-time. Feeding these estimated disturbances back into the parameters of the optimization solvers would improve trajectory tracking accuracy under severe environmental conditions.
- **Resilient Decentralized Communication:** Investigate the robustness of the decentralized coordinator under communication failures, packet loss, or latency in the USV-to-UAV data link, developing fail-safe behaviors where the vehicles fall back to local safety controllers.
- **Tether Power Delivery and Energy Harvesting:** Explore the physical optimization of power transfer through the tether to supply the UAV from the catamaran's main batteries, extending flight times indefinitely. Additionally, energy regeneration strategies could be investigated, utilizing the winch motor to harvest energy from the pulling forces exerted by the UAV during high-tension maneuvers.

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